Mechanics III

Study Document: Dynamics of Particles, Systems of Particles and Rigid Bodies

Typeset and expanded from handwritten lecture notes (Lectures 1–13)

Autumn 2026

How to use this document. Everything from the handwritten notes is reproduced here, in the original order, with the original example numbering. On top of that, every result is *derived* step by step, and short explanations are added where the notes only sketch an idea. Boxes are colour-coded:

- **blue** – definitions,
- **green** – key results / formulas to memorise,
- **orange** – worked examples,
- **grey** – remarks and intuition,
- **red** – common pitfalls.

Notation. Vectors are bold ($\boldsymbol{r}$; in handwriting/lecture they are underlined, $\underline{r}$), second-order tensors are double underlined ($\underline{\underline{I}}$), $[\cdot]_C$ denotes components in frame C, and a dot denotes a time derivative, $\dot{x} = \mathrm{d}x/\mathrm{d}t$. Abbreviations: **DOF** = degree(s) of freedom, **FBD** = free-body diagram, **LMB** = linear momentum balance, **AMB** = angular momentum balance, **WEB** = work–energy balance, **ICR** = instantaneous centre of rotation, **IAR** = instantaneous axis of rotation, cm = centre of mass.

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1 Single Particle Dynamics

Lecture 1 (16.09.2026)

Definition: Particle

A *particle* is an object of negligible size and shape – mathematically, a point carrying a mass m. Its rotation is irrelevant; only its position matters.

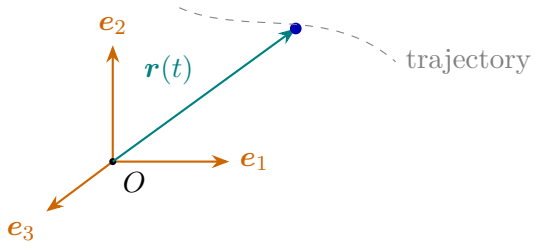
Kinematics is the description of motion (positions, velocities, accelerations) *without* asking what causes it. *Kinetics* (Section 2) then relates the motion to forces.

1.1 Kinematics in a Cartesian reference frame

Definition: Cartesian frame

A Cartesian frame C consists of

- a **fixed origin** O,
- a **fixed orthonormal basis** $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$, i.e. $\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij}$ and $\dot{\boldsymbol{e}}_i = \boldsymbol{0}$.



Position.

$$\boldsymbol{r}(t) = \sum_{i=1}^{3} x_i(t)\, \boldsymbol{e}_i, \tag{1.1}$$

where the scalar functions $x_i(t)$ are the *coordinates*. Each independent coordinate that is needed to describe the configuration is one *degree of freedom*.

Key result

A (free) particle in d dimensions has d degrees of freedom.

(Particle rotation does not matter, hence DOF $= d$ – compare rigid bodies in Section 4, which have additional rotational DOFs.)

Velocity. Because the basis vectors are *constant* in a Cartesian frame, differentiation acts only on the coordinates:

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \sum_{i=1}^{d} \frac{\mathrm{d}x_i}{\mathrm{d}t}\, \boldsymbol{e}_i = \sum_{i=1}^{d} \dot{x}_i\, \boldsymbol{e}_i. \tag{1.2}$$

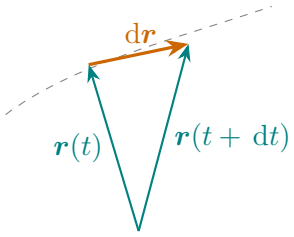
Acceleration.

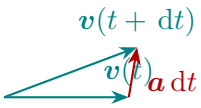
$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \sum_{i=1}^{d} \ddot{x}_i\, \boldsymbol{e}_i. \tag{1.3}$$

Geometric meaning. Over an infinitesimal time step $\mathrm{d}t$,

$$\mathrm{d}\boldsymbol{r} = \boldsymbol{v}\,\mathrm{d}t, \qquad \mathrm{d}\boldsymbol{v} = \boldsymbol{a}\,\mathrm{d}t. \tag{1.4}$$

So the velocity always points in the direction of $\mathrm{d}\boldsymbol{r}$, i.e. it is *tangent to the trajectory* ("where you are moving"), whereas the acceleration points in the direction of the *change* of velocity $\boldsymbol{v}(t+\mathrm{d}t)-\boldsymbol{v}(t)$. In particular the acceleration need *not* be tangent to the path.





Speed. The speed is the magnitude of the velocity,

$$v = |\boldsymbol{v}| = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}} = \sqrt{\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2}. \tag{1.5}$$

Component form. Collecting the coordinates in a column (with the frame as subscript),

$$[\boldsymbol{r}(t)]_C = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}, \qquad [\boldsymbol{v}(t)]_C = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix}, \qquad [\boldsymbol{a}(t)]_C = \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix}. \tag{1.6}$$

Remark

Vectors are frame-independent objects; their *components* depend on the chosen frame. That is why we write $[\boldsymbol{r}]_C$: the column is the representation of $\boldsymbol{r}$ in frame C.

Finding the trajectory from the acceleration

Example 1.1: Finding $\boldsymbol{r}(t)$ from a known $\boldsymbol{a}(t)$

Idea. Integrate twice. Start from the definition $\boldsymbol{a} = \mathrm{d}\boldsymbol{v}/\mathrm{d}t$ and separate:

$$\boldsymbol{a}(t) = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \iff \int_{t_0}^{t} \boldsymbol{a}(\tau)\,\mathrm{d}\tau = \int_{\boldsymbol{v}(t_0)}^{\boldsymbol{v}(t)} \mathrm{d}\boldsymbol{v} \implies \boxed{\boldsymbol{v}(t) = \boldsymbol{v}(t_0) + \int_{t_0}^{t} \boldsymbol{a}(\tau)\,\mathrm{d}\tau}.$$

(Using definite integrals with the initial condition as lower limit means we never need an "$+C$".) Analogously, from $\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$:

$$\boxed{\boldsymbol{r}(t) = \boldsymbol{r}(t_0) + \int_{t_0}^{t} \boldsymbol{v}(\tau)\,\mathrm{d}\tau}.$$

Application: free fall under gravity. Choose $\boldsymbol{e}_2$ pointing up, so $\boldsymbol{a}(t) = -g\,\boldsymbol{e}_2 = \text{const}$. Take $t_0 = 0$. Then

$$\boldsymbol{v}(t) = \boldsymbol{v}(0) - g\,t\,\boldsymbol{e}_2,$$
$$\boldsymbol{r}(t) = \boldsymbol{r}(0) + \int_0^t \left(\boldsymbol{v}(0) - g\tau\boldsymbol{e}_2\right)\mathrm{d}\tau = \boldsymbol{r}(0) + \boldsymbol{v}(0)\,t - \frac{gt^2}{2}\,\boldsymbol{e}_2,$$

or in components

$$\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \underbrace{\begin{pmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{pmatrix} + \begin{pmatrix} \dot{x}_1(0) \\ \dot{x}_2(0) \\ \dot{x}_3(0) \end{pmatrix} t}_{\text{initial conditions}} + \begin{pmatrix} 0 \\ -\frac{gt^2}{2} \\ 0 \end{pmatrix}.$$

A second-order ODE needs *two* vector initial conditions: $\boldsymbol{r}(0)$ and $\boldsymbol{v}(0)$.

Clicker question: who hits the ground first?

Two identical particles are launched at the same time. Particle 1 starts vertically upward from the ground with initial velocity v_0; particle 2 is released from height h with zero initial velocity. Both fall with constant acceleration g. From what height h must particle 2 be released so that particle 1 hits the ground first?

Particle 1. With $x_2^{(1)}(0) = 0$, $\dot{x}_2^{(1)}(0) = v_0$:

$$x_2^{(1)}(t) = v_0 t - \tfrac{1}{2}gt^2 = t\left(v_0 - \tfrac{1}{2}gt\right) \stackrel{!}{=} 0 \implies t_1 = 0 \text{ (launch)}, \quad t^{(1)} = \frac{2v_0}{g} \text{ (landing)}.$$

Particle 2. With $x_2^{(2)}(0) = h$, $\dot{x}_2^{(2)}(0) = 0$:

$$x_2^{(2)}(t) = h - \tfrac{1}{2}gt^2 \stackrel{!}{=} 0 \implies t^{(2)} = \sqrt{\frac{2h}{g}}.$$

Condition $t^{(1)} < t^{(2)}$ (both sides positive, so squaring is allowed):

$$\frac{2v_0}{g} < \sqrt{\frac{2h}{g}} \iff \frac{4v_0^2}{g^2} < \frac{2h}{g} \iff \boxed{h > \frac{2v_0^2}{g}}.$$

Sanity check: particle 1 reaches a maximum height $v_0^2/(2g)$; particle 2 must start four times higher.

Example 1.5: Finding $r(t)$ from a known $a(v)$ or $a(x)$

1) Drag (acceleration depends on velocity). A particle moving through a medium experiences a drag. For slow motion, *Stokes drag* is linear: $\boldsymbol{a} = -k\boldsymbol{v}$, $k > 0$. Consider 1D: $\boldsymbol{r} = x\boldsymbol{e}_1$, $\boldsymbol{v} = v\boldsymbol{e}_1$, $\boldsymbol{a} = a\boldsymbol{e}_1$ with $a = a(v) = -kv$.

Since a depends on v and not explicitly on t, separate variables:

$$a(v) = \frac{\mathrm{d}v}{\mathrm{d}t} \implies \int_{t_0}^{t} \mathrm{d}\tau = \int_{v(t_0)}^{v(t)} \frac{\mathrm{d}v}{a(v)} = \int_{v(t_0)}^{v(t)} \frac{\mathrm{d}v}{-kv} \implies t - t_0 = -\frac{1}{k} \ln \frac{v(t)}{v(t_0)}.$$

Solving for v:

$$\boxed{v(t) = v(t_0)\, e^{-k(t-t_0)}}.$$

The velocity decays exponentially; integrating once more gives $x(t) = x(t_0) + \frac{v(t_0)}{k}(1 - e^{-k(t-t_0)})$, so the particle stops after a finite distance $v(t_0)/k$.

2) Rocket launch (acceleration depends on position). Far from the earth, gravity depends on the altitude: $a = a(x)$. Use the chain rule to eliminate t:

$$a(x) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x} \underbrace{\frac{\mathrm{d}x}{\mathrm{d}t}}_{=v} = v \frac{\mathrm{d}v}{\mathrm{d}x}.$$

Separate and integrate:

$$\int_{v(x_0)}^{v(x)} v\,\mathrm{d}v = \int_{x_0}^{x} a(\xi)\,\mathrm{d}\xi \implies \boxed{\tfrac{1}{2}\big(v(x)^2 - v(x_0)^2\big) = \int_{x_0}^{x} a(\xi)\,\mathrm{d}\xi}.$$

This gives $v = v(x)$. To get the position as a function of time, separate once more: $v(x) = \mathrm{d}x/\mathrm{d}t \iff \mathrm{d}t = \mathrm{d}x/v(x)$, so $t - t_0 = \int_{x_0}^{x} \mathrm{d}\xi/v(\xi)$, which can be inverted to $x(t)$.
(The relation $\frac{1}{2}\,\mathrm{d}(v^2) = a\,\mathrm{d}x$ is the 1D precursor of the work–energy balance, Section 2.3.)

1.2 Constrained motion

Key result: Counting DOFs with constraints

If a particle in d dimensions is subject to k *independent* kinematic constraints, then it has

$$\mathrm{DOF} = d - k. \tag{1.7}$$

Example 1.6: (Un-)constrained motion

1. **Free motion:** $\boldsymbol{r} = \sum_{i=1}^{d} x_i \boldsymbol{e}_i \Rightarrow d$ DOFs.
2. **Particle on a rigid ground** (inclined plane with normal $\boldsymbol{n}$): the constraint $\boldsymbol{r}\cdot\boldsymbol{n} = \text{const}$ keeps the particle on the plane. In 3D: $3 - 1 = 2$ DOFs, $\boldsymbol{r} = x_1\boldsymbol{e}_1 + x_2\boldsymbol{e}_2$ with $\boldsymbol{e}_1, \boldsymbol{e}_2$ in the plane.
3. **Particle on a rigid link** of length L attached to a fixed point: $|\boldsymbol{r}| = L = \text{const} \Rightarrow d-1$ DOFs (a circle in 2D, a sphere in 3D).
4. **Particle in a tube / on a track:** only the path length s is free $\Rightarrow$ 1 DOF suffices.

Key result: Reaction forces

Every constraint gives rise to a **reaction force** – the force that "enforces" the constraint (normal force of the ground, tension in a link, ...). Moreover, whenever the velocity vector changes (in magnitude *or* direction) an acceleration exists, and hence (by Newton) a force must act.

1.3 Kinematics in polar coordinates (2D)

Lecture 2 (21.09.2026)

Describe the position by the distance $r(t)$ from the origin and the angle $\varphi(t)$ measured from $\boldsymbol{e}_1$:

$$\boldsymbol{r}(t) = \underbrace{r\cos\varphi}_{=x_1}\,\boldsymbol{e}_1 + \underbrace{r\sin\varphi}_{=x_2}\,\boldsymbol{e}_2. \tag{1.8}$$

Differentiating directly (product + chain rule) gives

$$\boldsymbol{v} = \dot{\boldsymbol{r}} = (\dot r\cos\varphi - r\sin\varphi\,\dot\varphi)\boldsymbol{e}_1 + (\dot r\sin\varphi + r\cos\varphi\,\dot\varphi)\boldsymbol{e}_2, \tag{1.9}$$

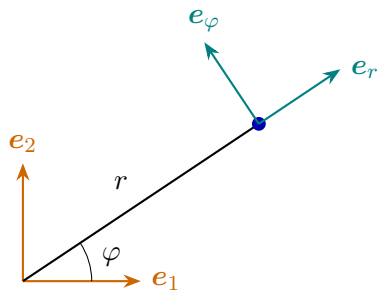
and the acceleration in this form becomes very complicated. Much cleaner: use a *co-rotating* basis.

Definition: Polar (co-rotating) basis

$$\boldsymbol{e}_r(t) = \cos\varphi\,\boldsymbol{e}_1 + \sin\varphi\,\boldsymbol{e}_2, \qquad \boldsymbol{e}_\varphi(t) = -\sin\varphi\,\boldsymbol{e}_1 + \cos\varphi\,\boldsymbol{e}_2. \tag{1.10}$$

It rotates with the particle; it is **time dependent** but still orthonormal: $|\boldsymbol{e}_r| = |\boldsymbol{e}_\varphi| = 1$,

$\boldsymbol{e}_r \cdot \boldsymbol{e}_\varphi = 0.$



Derivatives of the basis vectors. By the chain rule,

Key result

$$\dot{\boldsymbol{e}}_r = -\sin\varphi\,\dot\varphi\,\boldsymbol{e}_1 + \cos\varphi\,\dot\varphi\,\boldsymbol{e}_2 = \dot\varphi\,\boldsymbol{e}_\varphi, \tag{1.11}$$
$$\dot{\boldsymbol{e}}_\varphi = -\cos\varphi\,\dot\varphi\,\boldsymbol{e}_1 - \sin\varphi\,\dot\varphi\,\boldsymbol{e}_2 = -\dot\varphi\,\boldsymbol{e}_r. \tag{1.12}$$

Remark

These are the only two identities you need. Interpretation: a unit vector can only rotate, so its derivative is perpendicular to it, with magnitude equal to the rotation rate $\dot\varphi$.

Position. $\boldsymbol{r}(t) = r(t)\,\boldsymbol{e}_r(t).$

Velocity. Product rule:

$$\boldsymbol{v} = \dot{\boldsymbol{r}} = \dot r\,\boldsymbol{e}_r + r\,\dot{\boldsymbol{e}}_r = \boxed{\dot r\,\boldsymbol{e}_r + r\dot\varphi\,\boldsymbol{e}_\varphi}. \tag{1.13}$$

The radial part $\dot r$ is due to the change of distance ("radius change"), the circumferential part $r\dot\varphi$ is due to rotation.

Definition: Angular velocity / acceleration

$$\omega := \dot\varphi \;\; [\mathrm{rad\,s^{-1}}], \qquad \dot\omega = \ddot\varphi \;\; [\mathrm{rad\,s^{-2}}]. \tag{1.14}$$

Acceleration. Differentiate the velocity once more, again using the basis identities:

$$\begin{aligned}\boldsymbol{a} = \dot{\boldsymbol{v}} &= \ddot r\,\boldsymbol{e}_r + \dot r\,\dot{\boldsymbol{e}}_r + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + r\ddot\varphi\,\boldsymbol{e}_\varphi + r\dot\varphi\,\dot{\boldsymbol{e}}_\varphi \\ &= \ddot r\,\boldsymbol{e}_r + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + r\ddot\varphi\,\boldsymbol{e}_\varphi - r\dot\varphi^2\,\boldsymbol{e}_r \\ &= \boxed{(\ddot r - r\dot\varphi^2)\boldsymbol{e}_r + (2\dot r\dot\varphi + r\ddot\varphi)\boldsymbol{e}_\varphi}.\end{aligned} \tag{1.15}$$

- $-r\dot\varphi^2$: **centripetal acceleration** (always points towards the origin),
- $2\dot r\dot\varphi$: **Coriolis term** – if you move outwards on a rotating platform you have to accelerate sideways, because the circumferential speed $r\dot\varphi$ grows with r,
- $r\ddot\varphi$: tangential acceleration due to angular acceleration.

Common pitfall

$v_r = \dot{r}$, but $a_r \neq \dot{v}_r$! Because the basis is time dependent, the components of $\boldsymbol{a}$ are *not* the time derivatives of the components of $\boldsymbol{v}$. The same holds for the φ-parts.

Example 1.7: Particle on a taut, inextensible string

The particle moves on a circle of radius L: $r(t) = L = \text{const} \Rightarrow \dot{r} = \ddot{r} = 0$. Then

$$\boldsymbol{v}(t) = L\dot{\varphi}(t)\,\boldsymbol{e}_\varphi(t), \qquad |\boldsymbol{v}| = L|\dot{\varphi}| = L|\omega|,$$
$$\boldsymbol{a}(t) = -L\dot{\varphi}^2(t)\,\boldsymbol{e}_r(t) + L\ddot{\varphi}(t)\,\boldsymbol{e}_\varphi(t).$$

For uniform rotation, $\dot{\varphi} = \omega = \text{const}$:

$$\boldsymbol{v} = L\omega\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{a} = -L\omega^2\,\boldsymbol{e}_r = -\frac{v^2}{L}\,\boldsymbol{e}_r \quad \text{(centripetal acceleration)}.$$

Even at constant speed there is an acceleration, because the *direction* of $\boldsymbol{v}$ changes.

Clicker: walking radially on a rotating disk

A disk rotates with $\omega = \text{const}$; a particle moves radially *inwards* at constant speed v_0, so $\dot{r} = -v_0$, $\ddot{r} = 0$, $\dot{\varphi} = \omega$, $\ddot{\varphi} = 0$. Then

$$\boldsymbol{v} = \dot{r}\,\boldsymbol{e}_r + r\dot{\varphi}\,\boldsymbol{e}_\varphi = -v_0\,\boldsymbol{e}_r + r\omega\,\boldsymbol{e}_\varphi,$$
$$\boldsymbol{a} = -r\omega^2\,\boldsymbol{e}_r + 2(-v_0)\omega\,\boldsymbol{e}_\varphi,$$
$$|\boldsymbol{a}| = \sqrt{(r\omega^2)^2 + (2v_0\omega)^2}.$$

Since v_0 and ω are constant, $|\boldsymbol{a}|$ is minimal when $r = 0$ (centre of the disk): $|\boldsymbol{a}|_{\min} = 2|v_0\omega|$ – the Coriolis part never vanishes.

1.4 Kinematics of space curves (path coordinates)

If the path is known (a track, a road), parametrise the position by the *path length* $s = s(t)$ measured along the curve: $\boldsymbol{r} = \boldsymbol{r}(s)$.

Definition: Tangent and normal vectors

- Unit tangent: $\boldsymbol{e}_t = \dfrac{\boldsymbol{v}}{|\boldsymbol{v}|} = \dfrac{\boldsymbol{v}}{v} = \dfrac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}$.
- Speed: $v = |\boldsymbol{v}| = \dot{s} = \mathrm{d}s/\mathrm{d}t$.
- Unit (principal) normal $\boldsymbol{e}_n$: points towards the centre of the *osculating circle*, whose radius ρ is the local radius of curvature.

Velocity. $\boldsymbol{v} = v\,\boldsymbol{e}_t = \dot{s}\,\boldsymbol{e}_t$.

Acceleration. Product rule: $\boldsymbol{a} = \ddot{s}\,\boldsymbol{e}_t + \dot{s}\,\dot{\boldsymbol{e}}_t$. The tangent vector turns as we move along the curve; locally the curve is a circle of radius ρ, so the tangent turns by the angle $\mathrm{d}\theta = \mathrm{d}s/\rho$, giving $\frac{\mathrm{d}\boldsymbol{e}_t}{\mathrm{d}s} = \frac{1}{\rho}\boldsymbol{e}_n$ and therefore $\dot{\boldsymbol{e}}_t = \frac{\mathrm{d}\boldsymbol{e}_t}{\mathrm{d}s}\dot{s} = \frac{\dot{s}}{\rho}\boldsymbol{e}_n$. Hence

Key result: Path coordinates

$$\boldsymbol{v} = \dot{s}\,\boldsymbol{e}_t, \qquad \boldsymbol{a} = \underbrace{\ddot{s}\,\boldsymbol{e}_t}_{\text{along the path}} + \underbrace{\frac{v^2}{\rho}\,\boldsymbol{e}_n}_{\text{towards centre of curvature}}. \tag{1.16}$$

Special case: flat ground. $\rho \to \infty \Rightarrow v^2/\rho \to 0 \Rightarrow \boldsymbol{a} = \ddot{s}\,\boldsymbol{e}_t$.

Example 1.10: Circular motion

For a circle of radius R: $\rho = R = \text{const}$, path length $s(t) = R\varphi(t)$. So

$$\boldsymbol{v} = \dot{s}\,\boldsymbol{e}_t = R\dot{\varphi}\,\boldsymbol{e}_t, \qquad \boldsymbol{a} = \ddot{s}\,\boldsymbol{e}_t + \frac{v^2}{\rho}\boldsymbol{e}_n = R\ddot{\varphi}\,\boldsymbol{e}_t + \frac{v^2}{R}\boldsymbol{e}_n.$$

Compare with polar coordinates for $r = R = \text{const}$: $\boldsymbol{a} = R\ddot{\varphi}\,\boldsymbol{e}_\varphi - \frac{v^2}{R}\boldsymbol{e}_r$. The tangential terms agree ($\boldsymbol{e}_t = \boldsymbol{e}_\varphi$), and the normal terms agree because $\boldsymbol{e}_n$ always points towards the centre of the osculating circle while $\boldsymbol{e}_r$ points away from it: $\boldsymbol{e}_n = -\boldsymbol{e}_r$.

Remark: Practical application: track design

If a straight track is joined directly to a circular arc, the curvature $1/\rho$ jumps from 0 to $1/R$, so the normal acceleration v^2/ρ (and the force on the passengers) jumps *instantaneously*. A better design uses a transition curve whose radius changes smoothly, so that v^2/ρ grows gradually (e.g. a clothoid, where curvature grows linearly with s). In a looping, an "inverted teardrop" shape keeps v^2/ρ roughly constant.

2 Single Particle Kinetics

Lecture 3 (23.09.2026)

Definition: Linear momentum

$$\boldsymbol{p} = m\boldsymbol{v} \qquad \text{(German: "Impuls")}. \tag{2.1}$$

2.1 Linear momentum balance (Newton's axioms)

Key result: Linear momentum balance (LMB)

$$\sum_i \boldsymbol{F}_i = \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}), \tag{2.2}$$

where $\sum_i \boldsymbol{F}_i$ is the *net force* acting on the particle. This holds in an *inertial* frame.

Special cases.

- **Statics:** $\boldsymbol{v} = \boldsymbol{0} \Rightarrow \boldsymbol{p} = \boldsymbol{0} \Rightarrow \sum_i \boldsymbol{F}_i = \boldsymbol{0}$.
- **Zero net force:** $\sum_i \boldsymbol{F}_i = \boldsymbol{0} \Rightarrow \boldsymbol{p} = \text{const} \Rightarrow m\boldsymbol{v} = \text{const}$ (*Newton's first axiom*: uniform straight-line motion).

- **Constant mass:** $\dot{m} = 0 \Rightarrow \dot{\boldsymbol{p}} = m\dot{\boldsymbol{v}}$ and

$$\boxed{\sum_i \boldsymbol{F}_i = m\boldsymbol{a}} \qquad (\textit{Newton's second axiom}). \tag{2.3}$$

  The force determines the *change* of velocity, not the velocity itself.

- **Zero mass:** $m = 0 \Rightarrow \boldsymbol{p} = \boldsymbol{0} \Rightarrow \sum_i \boldsymbol{F}_i = \boldsymbol{0}$ – massless objects (ropes, light pulleys, links) must always be in force equilibrium, even when they accelerate.

Newton's third axiom ("actio = reactio"). If body A exerts a force $\boldsymbol{F}$ on body B, then B exerts $-\boldsymbol{F}$ on A. This is what makes *free-body diagrams* (FBDs) consistent: when we cut two bodies apart, the contact force appears on both cut faces with opposite signs.

FBD of a mass resting on a block

A small mass m sits on a block of mass M which rests on the ground (gravity g downwards, $\boldsymbol{e}_2$ up). Cut the system into its parts:

- mass m: weight $-mg\,\boldsymbol{e}_2$, contact force from the block $N_1\boldsymbol{e}_2$;
- block M: weight $-Mg\,\boldsymbol{e}_2$, contact force from the mass $-N_1\boldsymbol{e}_2$ (Newton 3), ground force $N_2\boldsymbol{e}_2$ (really a distributed pressure, but we only care about its resultant).

LMB in $\boldsymbol{e}_2$:

$$N_1 - mg = ma_1, \qquad N_2 - N_1 - Mg = Ma_2.$$

Example 1.12: Block sliding on frictionless ground

Kinematics. The block can only move horizontally: $\boldsymbol{r}(t) = x_1(t)\boldsymbol{e}_1$, $\boldsymbol{v} = \dot{x}_1\boldsymbol{e}_1$, $\boldsymbol{a} = \ddot{x}_1\boldsymbol{e}_1$.
Friction. Friction coefficient $\mu = 0 \Rightarrow$ no friction force.
Kinetics (LMB).

$$\sum_i \boldsymbol{F}_i = (N - mg)\boldsymbol{e}_2 = m\boldsymbol{a} = m\ddot{x}_1\boldsymbol{e}_1.$$

The left side only has an $\boldsymbol{e}_2$-component, the right side only an $\boldsymbol{e}_1$-component, so both must vanish:

$$N = mg, \qquad \ddot{x}_1 = 0 \Rightarrow \boldsymbol{v} = \text{const}.$$

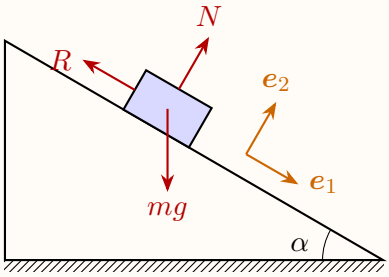
Coulomb friction.

- **Static friction** (no relative sliding): the friction force $\boldsymbol{R}$ adjusts itself, with $|\boldsymbol{R}| \le \mu_0|\boldsymbol{N}|$ (μ_0: static friction coefficient).

- **Kinetic (dynamic) friction** (sliding): magnitude fixed, direction opposite to the sliding velocity,

$$\boxed{\boldsymbol{R} = -\mu|\boldsymbol{N}|\frac{\boldsymbol{v}}{|\boldsymbol{v}|}} \qquad (\mu: \text{ dynamic friction coefficient}). \tag{2.4}$$

Example 1.13: Sliding down a rough incline



Coordinates: $\boldsymbol{e}_1$ down the slope, $\boldsymbol{e}_2$ normal to it.
Kinematics. $\boldsymbol{r} = x_1\boldsymbol{e}_1$, $\boldsymbol{v} = \dot{x}_1\boldsymbol{e}_1$, $\boldsymbol{a} = \ddot{x}_1\boldsymbol{e}_1$ (the block stays on the slope).
Kinetics. $\mu > 0 \Rightarrow$ friction. The block slides ($\dot{x}_1 > 0$), so kinetic friction acts up the slope: $\boldsymbol{R} = -R\boldsymbol{e}_1$ with $R = \mu N$.
LMB. Gravity mg decomposes into $mg\sin\alpha$ along $\boldsymbol{e}_1$ and $-mg\cos\alpha$ along $\boldsymbol{e}_2$:

$$\sum_i \boldsymbol{F}_i = (mg\sin\alpha - R)\boldsymbol{e}_1 + (N - mg\cos\alpha)\boldsymbol{e}_2 = m\ddot{x}_1\boldsymbol{e}_1.$$

Compare components:

$$\begin{aligned} \boldsymbol{e}_2: &\quad N = mg\cos\alpha, \\ \boldsymbol{e}_1: &\quad mg\sin\alpha - R = m\ddot{x}_1. \end{aligned}$$

With $R = \mu N = \mu mg\cos\alpha$:

$$\boxed{\ddot{x}_1 = g(\sin\alpha - \mu\cos\alpha)}.$$

If $\tan\alpha < \mu$ then $\ddot{x}_1 < 0$: the block is braking (decelerating) and eventually stops.

Example 1.14: Projectile motion

Only gravity acts: LMB $m\boldsymbol{a} = m\boldsymbol{g} \Rightarrow \boldsymbol{a} = \boldsymbol{g}$, independent of the mass. Hence the horizontal velocity stays exactly the same throughout the flight, and the vertical motion is the free fall of Example 1.1. The trajectory is a parabola.

Example 1.15: Particle on a string (uniform circular motion)

A mass m on a string of constant length L rotates around a fixed point. From polar kinematics (Example 1.7),

$$\text{LMB:} \quad \sum_i \boldsymbol{F}_i = m\boldsymbol{a} = m(-L\omega^2\boldsymbol{e}_r + L\dot{\omega}\,\boldsymbol{e}_\varphi).$$

With $\omega = \text{const}$ the only force (ignoring gravity) is the string force S pulling towards the centre:

$$-S\,\boldsymbol{e}_r = -mL\omega^2\,\boldsymbol{e}_r \quad \Longrightarrow \quad \boxed{S = mL\omega^2}.$$

2.2 Balance of angular momentum

Definition: Angular momentum with respect to a reference point B

$$\boldsymbol{H}_B = \boldsymbol{r}_{BP} \times \boldsymbol{p}, \qquad \boldsymbol{p} = m\boldsymbol{v}_P, \tag{2.5}$$

where $\boldsymbol{r}_{BP} = \boldsymbol{r}_P - \boldsymbol{r}_B$ points from the reference point B to the particle P. (German: "Drall" / "Drehimpuls".)

Derivation of the AMB. Differentiate with the product rule:

$$\dot{\boldsymbol{H}}_B = \dot{\boldsymbol{r}}_{BP} \times \boldsymbol{p} + \boldsymbol{r}_{BP} \times \dot{\boldsymbol{p}}.$$

- First term: $\dot{\boldsymbol{r}}_{BP} = \boldsymbol{v}_P - \boldsymbol{v}_B$, and $\boldsymbol{v}_P \times \boldsymbol{p} = \boldsymbol{v}_P \times m\boldsymbol{v}_P = \boldsymbol{0}$ (parallel vectors), so $\dot{\boldsymbol{r}}_{BP} \times \boldsymbol{p} = -\boldsymbol{v}_B \times \boldsymbol{p}$.
- Second term: by LMB $\dot{\boldsymbol{p}} = \sum_i \boldsymbol{F}_i$, so $\boldsymbol{r}_{BP} \times \sum_i \boldsymbol{F}_i =: \boldsymbol{M}_B$ is the *net torque* (moment) about B.

Thus $\dot{\boldsymbol{H}}_B = -\boldsymbol{v}_B \times \boldsymbol{p} + \boldsymbol{M}_B$, or

Key result: Angular momentum balance (AMB)

$$\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{p}. \tag{2.6}$$

For a single particle the AMB contains no new information beyond the LMB ($\sum \boldsymbol{F}_i = \dot{\boldsymbol{p}}$) – it was derived from it – but it is often much more convenient, because reaction forces passing through B drop out of $\boldsymbol{M}_B$.

Special cases.

- **Statics:** $\boldsymbol{H}_B = \boldsymbol{0}$, $\boldsymbol{p} = \boldsymbol{0} \Rightarrow \boldsymbol{M}_B = \boldsymbol{0}$ (makes sense, as the particle does not move).
- **Massless particle:** $\boldsymbol{p} = \boldsymbol{0}$, $\boldsymbol{H}_B = \boldsymbol{0} \Rightarrow \boldsymbol{M}_B = \boldsymbol{0}$ (as in statics).
- B **fixed** ($\boldsymbol{v}_B = \boldsymbol{0}$) **or** B **moving parallel to the particle** ($\boldsymbol{v}_B \parallel \boldsymbol{v}_P$): $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B$. *Choose B like this whenever possible!*
- **No torque** about B (and $\boldsymbol{v}_B \times \boldsymbol{p} = \boldsymbol{0}$): $\dot{\boldsymbol{H}}_B = \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{r}_{BP} \times m\boldsymbol{v}_P) = \boldsymbol{0}$ – angular momentum is conserved.

Special case: 2D, in polar coordinates around a fixed point B. With $\boldsymbol{r}_{BP} = r\,\boldsymbol{e}_r$ and $\boldsymbol{v}_P = v_r\boldsymbol{e}_r + v_\varphi\boldsymbol{e}_\varphi$, and $\boldsymbol{e}_r \times \boldsymbol{e}_r = \boldsymbol{0}$, $\boldsymbol{e}_r \times \boldsymbol{e}_\varphi = \boldsymbol{e}_3$:

$$\boldsymbol{H}_B = m\,r\boldsymbol{e}_r \times (v_r\boldsymbol{e}_r + v_\varphi\boldsymbol{e}_\varphi) = m\,r\,v_\varphi\,\boldsymbol{e}_3 = mr^2\omega\,\boldsymbol{e}_3.$$

Writing $\boldsymbol{M}_B = M_B\boldsymbol{e}_3$, the AMB becomes the scalar equation

$$\boxed{M_B = \frac{\mathrm{d}}{\mathrm{d}t}(mr^2\omega)} \qquad \text{(AMB in 2D with fixed } B\text{)}. \tag{2.7}$$

If additionally $r(t) = R = \text{const}$ (and $m = \text{const}$):

$$M_B = mR^2\dot{\omega} = \underbrace{mR^2}_{I_B}\ddot{\varphi}. \tag{2.8}$$

$I_B = mR^2$ is the *mass moment of inertia* – the rotational analogue of the mass in the LMB ($F = ma \;\leftrightarrow\; M_B = I_B\ddot{\varphi}$): it measures the resistance against angular acceleration.

2.3 Work–energy balance

Lecture 4 (28.09.2026)

Derivation. Start from the LMB, which always holds, and integrate it along the path from $\boldsymbol{r}_1 = \boldsymbol{r}(t_1)$ to $\boldsymbol{r}_2 = \boldsymbol{r}(t_2)$ (assume $m = \text{const}$):

$$\int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}) \cdot \mathrm{d}\boldsymbol{r} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r} =: W_{12}.$$

On the left use $\mathrm{d}\boldsymbol{r} = \boldsymbol{v}\,\mathrm{d}t$:

$$m\int_{t_1}^{t_2} \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \cdot \boldsymbol{v}\,\mathrm{d}t = m\int_{\boldsymbol{v}(t_1)}^{\boldsymbol{v}(t_2)} \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{v} = \frac{m|\boldsymbol{v}(t_2)|^2}{2} - \frac{m|\boldsymbol{v}(t_1)|^2}{2}.$$

Key result: Work–energy balance (WEB)

$$T(t_2) - T(t_1) = W_{12}, \qquad T = \frac{m|\boldsymbol{v}|^2}{2}, \qquad W_{12} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r}. \tag{2.9}$$

T is the *kinetic energy*, W_{12} the *work* done on the particle from 1 to 2: *the change in kinetic energy equals the work done.*

Remark

Forces perpendicular to the motion ($\boldsymbol{F} \perp \mathrm{d}\boldsymbol{r}$) do no work – e.g. the normal force on a fixed surface, or the string force on a pendulum. That is why the WEB is so useful: such reaction forces simply disappear.

Example 1.21: Frictional sliding – where does the particle stop?

A block with initial velocity v_0 slides on rough ground (μ). Kinematics: $\boldsymbol{r} = x_1\boldsymbol{e}_1$.
Kinetic energy: $T(t_1) = \frac{1}{2}mv_0^2$, $T(t_2) = 0$ (at rest).
Work: the forces are $(N - mg)\boldsymbol{e}_2$ and $-R\boldsymbol{e}_1$ with $N = mg$, $R = \mu N$. Since $\mathrm{d}\boldsymbol{r} = \mathrm{d}x_1\boldsymbol{e}_1$, only the friction force does work (forces perpendicular to the motion do no work):

$$W_{12} = \int_{x_1(t_1)}^{x_1(t_2)} (-\mu mg)\,\mathrm{d}x_1 = -\mu mg\,\Delta x.$$

WEB:

$$0 - \tfrac{1}{2}mv_0^2 = -\mu mg\,\Delta x \quad \Longrightarrow \quad \boxed{\Delta x = \frac{v_0^2}{2\mu g}}.$$

Note how we never needed the time.

Shortcut: conservative forces.

Definition: Conservative force, potential energy

A force $\boldsymbol{F}$ is *conservative* if there exists a potential $V(\boldsymbol{r})$ such that

$$\boldsymbol{F} = -\frac{\mathrm{d}V}{\mathrm{d}\boldsymbol{r}} \quad (= -\nabla V). \tag{2.10}$$

$V(\boldsymbol{r})$ is called the *potential energy.*

Example: gravity with $\boldsymbol{e}_2$ up, $V(\boldsymbol{r}) = mg\,\boldsymbol{r} \cdot \boldsymbol{e}_2 = mgx_2 \Rightarrow \boldsymbol{F} = -\nabla V = -mg\boldsymbol{e}_2$. ✓

For a conservative force the work is path independent:

$$W_{12} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} -\frac{\mathrm{d}V}{\mathrm{d}\boldsymbol{r}} \cdot \mathrm{d}\boldsymbol{r} = -\int_{V(\boldsymbol{r}_1)}^{V(\boldsymbol{r}_2)} \mathrm{d}V = \boxed{V(\boldsymbol{r}_1) - V(\boldsymbol{r}_2)}. \tag{2.11}$$

Check with a particle sliding along a frictionless curved track from height h_1 to h_2: the normal force is $\perp \boldsymbol{v}$ and does no work, so only $-mg\boldsymbol{e}_2$ counts: $W_{12} = \int -mg\boldsymbol{e}_2 \cdot \mathrm{d}\boldsymbol{r} = -mg \int_{x_2(t_1)}^{x_2(t_2)} \mathrm{d}x_2 = -mg(h_2 - h_1) = V_1 - V_2$. ✓

Reformulated WEB. Split the forces into conservative ones (with potential V) and non-conservative ones:

$$W_{12} = V(\boldsymbol{r}_1) - V(\boldsymbol{r}_2) + W_{12}^{\text{non-cons.}}, \qquad W_{12}^{\text{non-cons.}} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i^{\text{non-cons.}} \cdot \mathrm{d}\boldsymbol{r}.$$

Inserting into the WEB:

Key result: WEB with potential energy

$$T(t_2) + V(\boldsymbol{r}_2) - \big[T(t_1) + V(\boldsymbol{r}_1)\big] = W_{12}^{\text{non-cons.}}. \tag{2.12}$$

Definition: Conservative system

A system is conservative if work is done *only* by conservative forces. Then $W_{12}^{\text{non-cons.}} = 0$ and

$$T(t_2) + V(\boldsymbol{r}_2) = T(t_1) + V(\boldsymbol{r}_1) \quad \Longrightarrow \quad \boxed{T + V = \text{const}} \tag{2.13}$$

(conservation of mechanical energy: $E_{\text{kin}} + E_{\text{pot}} = \text{const}$).

Example 1.22: Maximum flight height

A projectile is launched with speed v_0 at angle α; no air drag. Gravity is conservative.
At launch (1): $T_1 = \frac{m}{2}v_0^2$, $V_1 = 0$. At the top (2): the vertical velocity vanishes, while the horizontal velocity is unchanged (Example 1.14), so $T_2 = \frac{m}{2}(v_0 \cos\alpha)^2$, $V_2 = mg\, x_{2,\max}$.
Energy conservation:

$$\frac{m}{2}v_0^2 + 0 = \frac{m}{2}(v_0\cos\alpha)^2 + mg\, x_{2,\max} \implies x_{2,\max} = \frac{v_0^2}{2g}(1 - \cos^2\alpha) = \boxed{\frac{v_0^2}{2g}\sin^2\alpha}.$$

Example 1.23: Car in a looping

A car (particle m) starts from rest at height h_0 and rolls without friction ($\mu = 0$, no engine, only gravity) down a ramp into a circular looping of radius R. Questions: (1) minimal h_0 to complete the looping; (2) the maximum force during the looping.
(1) Velocity from the WEB. Only gravity does work, so energy is conserved:

$$mgh_0 = mgh + \tfrac{1}{2}mv^2 \quad \Longrightarrow \quad v^2 = 2g(h_0 - h) \geq 0.$$

This only gives the *necessary* condition $h_0 \geq h_{\max} = 2R$. It is **not sufficient**: with h_0 slightly above $2R$ the car would still have speed at the top, but it falls off the track *before* reaching it – it must also be pressed against the track.
Sufficient condition: $N(\varphi) \geq 0$ (the car stays on the track). Let φ be measured from the bottom (entrance) of the looping, $\boldsymbol{e}_r$ pointing outwards from the centre, $h = R - R\cos\varphi$. LMB in the $\boldsymbol{e}_r$ direction (normal force pushes inwards, gravity component $mg\cos\varphi$ outwards,

acceleration $a_r = -R\dot{\varphi}^2 = -v^2/R$):

$$-N(\varphi) + mg\cos\varphi = -mR\dot{\varphi}^2 = -m\frac{v^2}{R} \implies N(\varphi) = mg\cos\varphi + \frac{mv^2}{R}.$$

Insert $v^2 = 2g(h_0 - h)$ and $h = R(1-\cos\varphi)$:

$$N(\varphi) = mg\cos\varphi + \frac{2mg(h_0 - R + R\cos\varphi)}{R} = mg\left(3\cos\varphi - 2 + \frac{2h_0}{R}\right).$$

Worst case: top of the looping, $\varphi = \pi$ ($\cos\varphi = -1$):

$$N(\pi) = mg\left(-3 - 2 + \frac{2h_0}{R}\right) \geq 0 \quad \implies \quad \boxed{h_0 \geq \frac{5R}{2}}.$$

(2) Maximum force at the bottom, $\varphi = 0$ ($\cos\varphi = 1$), for the minimal $h_0 = 5R/2$:

$$N_{\max} = N(0) = mg\,(3 - 2 + 5) = \boxed{6mg}.$$

Passengers feel six times their weight at the entrance of a circular looping – the reason real loopings are not circular.

Lecture 5 (30.09.2026)

Example 1.17: Pendulum – three ways to get the equation of motion

A mass m on a massless string of length R, pivot B fixed, angle φ from the downward vertical. Question: what is $\ddot{\varphi}$?

FBD. String force F (towards B, i.e. $-F\boldsymbol{e}_r$) and weight mg (downwards). In polar coordinates around B (with $\boldsymbol{e}_r$ pointing from B to the mass), gravity has components $mg\cos\varphi\,\boldsymbol{e}_r - mg\sin\varphi\,\boldsymbol{e}_\varphi$.

(a) LMB using the polar acceleration with $r = R = \text{const}$:

$$\sum \boldsymbol{F}_i = (-F + mg\cos\varphi)\boldsymbol{e}_r - mg\sin\varphi\,\boldsymbol{e}_\varphi = m(-R\dot{\varphi}^2\boldsymbol{e}_r + R\ddot{\varphi}\,\boldsymbol{e}_\varphi).$$

Comparing coefficients:

$$\boldsymbol{e}_\varphi: \quad mR\ddot{\varphi} = -mg\sin\varphi \;\Rightarrow\; \boxed{\ddot{\varphi} = -\frac{g}{R}\sin\varphi},$$
$$\boldsymbol{e}_r: \quad F = mg\cos\varphi + mR\dot{\varphi}^2.$$

(Same structure as the normal force in the looping: $mR\dot{\varphi}^2 = mv^2/R$.)

(b) AMB about the fixed point B (the string force passes through B, so it has no moment; gravity has lever arm $R\sin\varphi$):

$$M_B = -mgR\sin\varphi \overset{!}{=} mR^2\ddot{\varphi} \;\Rightarrow\; \ddot{\varphi} = -\frac{g}{R}\sin\varphi.$$

The unknown string force never appears – this is the advantage of choosing B at the pivot.

(c) WEB. The system is conservative (no drag; $F \perp \boldsymbol{v}$ does no work), so $T + V = \text{const}$. With $V = 0$ at the pivot height and release from rest at a height $-h_0$:

$$-mgh_0 = -mg\underbrace{R\cos\varphi}_{h(\varphi)} + \frac{m}{2}\underbrace{R^2\dot{\varphi}^2}_{v^2}.$$

Differentiate with respect to time (the left side is constant):

$$0 = mgR\sin\varphi\,\dot{\varphi} + \frac{m}{2}R^2\,2\dot{\varphi}\ddot{\varphi} \implies g\sin\varphi + R\ddot{\varphi} = 0 \implies \ddot{\varphi} = -\frac{g}{R}\sin\varphi.$$

For small angles ($\sin\varphi \approx \varphi$) this is a harmonic oscillator with $\omega_0 = \sqrt{g/R}$.

2.4 Particle impact on a rigid, smooth wall

The wall is rigid (does not move, not deformable) and smooth (frictionless, so it only pushes along its normal $\boldsymbol{e}_n$). During the very short impact the wall force $\boldsymbol{F}(t)$ is huge.

Impulse. Integrate the LMB over the impact interval $[t_-, t_+]$:

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}) = \boldsymbol{F}(t) \implies \int_{m\boldsymbol{v}(t_-)}^{m\boldsymbol{v}(t_+)} \mathrm{d}(m\boldsymbol{v}) = \int_{t_-}^{t_+} \boldsymbol{F}(t)\,\mathrm{d}t.$$

Definition: Impulsive force / impulse

$$m\boldsymbol{v}(t_+) - m\boldsymbol{v}(t_-) = \int_{t_-}^{t_+} \boldsymbol{F}(t)\,\mathrm{d}t =: \hat{\boldsymbol{F}} \qquad [\mathrm{N\,s}], \tag{2.14}$$

the total momentum transferred to the particle during the impact.

Two phases. Split the impact at the instant t_0 of maximum compression:

- **compression** $[t_-, t_0]$: the particle is slowed down and deformed, $\hat{\boldsymbol{F}}_{\text{comp}} = \int_{t_-}^{t_0} \boldsymbol{F}\,\mathrm{d}t$;
- **restitution** $[t_0, t_+]$: the particle (partially) returns to its original shape, $\hat{\boldsymbol{F}}_{\text{rest}} = \int_{t_0}^{t_+} \boldsymbol{F}\,\mathrm{d}t$.

Since the wall is smooth, all impulses act along $\boldsymbol{e}_n$: $\hat{\boldsymbol{F}}_i = \hat{F}_i\,\boldsymbol{e}_n$.

Decompose $\boldsymbol{v} = v_n\boldsymbol{e}_n + v_t\boldsymbol{e}_t$. LMB per phase:

$$m\boldsymbol{v}(t_0) - m\boldsymbol{v}(t_-) = \hat{F}_{\text{comp}}\boldsymbol{e}_n,$$
$$m\boldsymbol{v}(t_+) - m\boldsymbol{v}(t_0) = \hat{F}_{\text{rest}}\boldsymbol{e}_n.$$

Tangential part: no tangential impulse, so

$$\boxed{v_t(t_-) = v_t(t_0) = v_t(t_+)}. \tag{2.15}$$

Normal part: at maximum compression the normal velocity vanishes, $v_n(t_0) = 0$, hence

$$-mv_n(t_-) = \hat{F}_{\text{comp}}, \qquad mv_n(t_+) = \hat{F}_{\text{rest}}.$$

Two equations, but three unknowns ($v_n(t_+)$, $\hat{F}_{\text{comp}}$, $\hat{F}_{\text{rest}}$) – we need a material law.

Definition: Coefficient of restitution

$$e = \frac{\hat{F}_{\text{rest}}}{\hat{F}_{\text{comp}}} \in [0, 1]. \tag{2.16}$$

- $e = 1$: (perfectly) elastic – full recovery, $\hat{F}_{\text{comp}} = \hat{F}_{\text{rest}}$;
- $e = 0$: (perfectly) plastic impact – no recovery, $\hat{F}_{\text{rest}} = 0$;
- $e \in (0, 1)$: partially elastic – partial recovery.

Combining:

$$-\frac{v_n(t_+)}{v_n(t_-)} = \frac{\hat{F}_{\text{rest}}}{\hat{F}_{\text{comp}}} = e \quad \implies \quad \boxed{v_n(t_+) = -e\,v_n(t_-)}.$$

(The sign flips because the particle bounces back.) So: $e = 1$ – normal velocity just flips; $e = 0$ – the particle stays at the wall, sliding along it with unchanged v_t; $e \in (0, 1)$ – v_t stays, $|v_n|$ becomes $e|v_n|$.

Remark: Why $e \leq 1$

$$T(t_-) = \frac{m}{2}(v_n^2 + v_t^2), \qquad T(t_+) = \frac{m}{2}((ev_n)^2 + v_t^2).$$

$e > 1$ would mean $T(t_+) > T(t_-)$: energy created from nothing – physically impossible for a passive wall.

Example 1.26: Measuring e – particle dropped onto the ground

Method 1 (rebound height). Drop from rest at height h_-; it rebounds to h_+. Energy is conserved during the free flights (the WEB does not hold *across* the collision, where energy is lost):

$$mgh_- = \frac{m}{2}v_n^2(t_-), \qquad mgh_+ = \frac{m}{2}v_n^2(t_+) \implies \frac{h_+}{h_-} = \frac{v_n^2(t_+)}{v_n^2(t_-)} = e^2 \implies \boxed{e = \sqrt{\frac{h_+}{h_-}}}.$$

Method 2 (flight times). Measure the time between subsequent impacts. After a bounce with upward speed v_0, LMB: $-mg = m\ddot{x}_2 \Rightarrow x_2(t) = -\frac{g}{2}t^2 + v_0 t = t\left(v_0 - \frac{gt}{2}\right)$, so the time of flight is $t^* = 2v_0/g$. For two consecutive impacts:

$$t_1^* = \frac{2v_0}{g}, \qquad t_2^* = \frac{2ev_0}{g} \implies \boxed{e = \frac{t_2^*}{t_1^*}}.$$

3 Dynamics of Systems of Particles

Lecture 6 (05.10.2026)

3.1 Kinematics

Consider n particles with masses m_i and positions $\boldsymbol{r}_i$.

Definition: Total mass and centre of mass

$$M = \sum_{i=1}^{n} m_i, \qquad \boldsymbol{r}_{\text{cm}} = \frac{1}{M}\sum_{i=1}^{n} m_i \boldsymbol{r}_i. \tag{3.1}$$

Definition: Holonomic constraints

A kinematic constraint is *holonomic* if it can be written as an equation between positions (and time):

$$f(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_n, t) = 0. \tag{3.2}$$

Examples:

- rigid link between i and j (cannot change its length): $|\boldsymbol{r}_i(t) - \boldsymbol{r}_j(t)| = \text{const}$;
- rigid system: $|\boldsymbol{r}_i - \boldsymbol{r}_j| = \text{const}$ for all pairs i, j of the system.

Key result: DOFs of a system of particles

A system of n particles in d dimensions with r independent holonomic constraints has

$$\text{DOF} = nd - r. \tag{3.3}$$

(Each particle contributes d DOFs; each constraint removes one.)

Example 2.1: Counting DOFs

1. **Two particles joined by a rigid link in 3D:** $2 \cdot 3 - 1 = 5$ DOFs (3 for the position of one particle, 2 angles for the direction of the link).
2. **Two masses on ropes wound around a stepped pulley** (radii r_0, r_1, rotation φ): $x_0 = r_0\varphi$, $x_1 = r_1\varphi \Rightarrow x_0/x_1 = r_0/r_1$; everything is determined by φ: 1 DOF (for the up–down motion).
3. **Four particles forming a quadrilateral in 2D**, connected by four rigid links (a diagonal spring is *not* a constraint), one corner pinned, one on a roller: $4 \cdot 2 = 8$ DOFs; pin -2, roller -1, rigid links $-4 \Rightarrow 8 - 7 = 1$ DOF in total.

3.2 Kinetics

Internal and external forces. On particle i act an external force $\boldsymbol{F}_{i,\text{ext}}$ and internal forces $\boldsymbol{F}_{ij}$ from every other particle j (e.g. via massless connections). By Newton's third axiom, $\boldsymbol{F}_{ij} = -\boldsymbol{F}_{ji}$ (and they act along the connecting line). The net force on particle i is

$$\boldsymbol{F}_i = \boldsymbol{F}_{i,\text{ext}} + \sum_{j \neq i} \boldsymbol{F}_{ij}.$$

Summing over all particles, the internal forces cancel pairwise (for every force there is a reaction):

$$\boxed{\sum_{i=1}^{n} \boldsymbol{F}_i = \sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}} + \boldsymbol{0}} \qquad \text{(sum of all forces = sum of external forces).} \tag{3.4}$$

3.2.1 Balance of linear momentum

LMB for each particle: $\boldsymbol{F}_i = \frac{\mathrm{d}}{\mathrm{d}t}(m_i \boldsymbol{v}_i)$. Summing,

$$\sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}} = \sum_{i=1}^{n} \frac{\mathrm{d}}{\mathrm{d}t}(m_i \boldsymbol{v}_i) = \frac{\mathrm{d}}{\mathrm{d}t}\Big(M \cdot \underbrace{\frac{1}{M}\sum_{i=1}^{n} m_i \dot{\boldsymbol{r}}_i}_{=\dot{\boldsymbol{r}}_{\text{cm}}}\Big) = \frac{\mathrm{d}}{\mathrm{d}t}(M \boldsymbol{v}_{\text{cm}}).$$

Key result: LMB for a system of particles

$$\boldsymbol{P} = M\boldsymbol{v}_{\text{cm}}, \qquad \sum_{i=1}^{n} \boldsymbol{F}_i^{\text{ext}} = \frac{\mathrm{d}}{\mathrm{d}t}(M\boldsymbol{v}_{\text{cm}}) \overset{M=\text{const}}{\Longrightarrow} \sum_{i=1}^{n} \boldsymbol{F}_i^{\text{ext}} = M\boldsymbol{a}_{\text{cm}}. \tag{3.5}$$

The centre of mass moves like a single particle of mass M under the total external force.

Example 2.4: Falling dumbbell

Two masses m_1, m_2 joined by a rigid massless link fall under gravity.
LMB on the system: only gravity is external, $m_1\boldsymbol{g} + m_2\boldsymbol{g} = (m_1 + m_2)\boldsymbol{a}_{\text{cm}} \Rightarrow \boldsymbol{a}_{\text{cm}} = \boldsymbol{g} \Rightarrow \boldsymbol{r}_{\text{cm}}(t) = \frac{1}{2}\boldsymbol{g}t^2 + \boldsymbol{v}_{\text{cm}}(0)t + \boldsymbol{r}_{\text{cm}}(0)$: the centre of mass flies on a parabola, no matter how the dumbbell tumbles.
LMB on each particle (the link forces appear, with $\boldsymbol{F}_{12} = -\boldsymbol{F}_{21}$):

$$m_1\boldsymbol{a}_1 = m_1\boldsymbol{g} + \boldsymbol{F}_{12}, \qquad m_2\boldsymbol{a}_2 = m_2\boldsymbol{g} + \boldsymbol{F}_{21}.$$

3.2.2 Work–energy balance for a system of particles

Summing the single-particle WEB $T_i(t_2) - T_i(t_1) = W_{i,12}$ over all particles:

Key result

$$T(t_2) - T(t_1) = W_{12}, \qquad T(t) = \frac{1}{2}\sum_{i=1}^{n} m_i |\boldsymbol{v}_i(t)|^2, \qquad W_{12} = \sum_{i=1}^{n} \int_{\boldsymbol{r}_i(t_1)}^{\boldsymbol{r}_i(t_2)} \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r}_i. \tag{3.6}$$

Common pitfall

Unlike in the LMB, *internal forces can do work* (e.g. a spring between two particles), so $\boldsymbol{F}_i$ here contains internal forces too.

Special cases.

- Conservative forces: $W_{12} = V_1 - V_2$. Conservative system: $T + V = \text{const}$.
- **A rigid link never performs work.** Proof: from $|\boldsymbol{r}_i - \boldsymbol{r}_j|^2 = \text{const}$, differentiating,

  $$2(\boldsymbol{r}_i - \boldsymbol{r}_j) \cdot (\dot{\boldsymbol{r}}_i - \dot{\boldsymbol{r}}_j) = 0.$$

  The link forces act along the link, $\boldsymbol{F}_{ij} \parallel \boldsymbol{r}_i - \boldsymbol{r}_j$, and $\boldsymbol{F}_{ji} = -\boldsymbol{F}_{ij}$, so their combined work is

  $$\int \boldsymbol{F}_{ij} \cdot \mathrm{d}\boldsymbol{r}_i + \int \boldsymbol{F}_{ji} \cdot \mathrm{d}\boldsymbol{r}_j = \int_{t_1}^{t_2} \boldsymbol{F}_{ij} \cdot (\boldsymbol{v}_i - \boldsymbol{v}_j)\, \mathrm{d}t = 0,$$

  since $\boldsymbol{F}_{ij} \parallel (\boldsymbol{r}_i - \boldsymbol{r}_j) \perp (\boldsymbol{v}_i - \boldsymbol{v}_j)$.

3.2.3 Balance of angular momentum

For a system, $\boldsymbol{H}_B = \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times \boldsymbol{p}_i$. Differentiate:

$$\dot{\boldsymbol{H}}_B = \sum_{i=1}^{n} \underbrace{\dot{\boldsymbol{r}}_{BP_i}}_{=\boldsymbol{v}_{P_i} - \boldsymbol{v}_B} \times \boldsymbol{p}_i + \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times \underbrace{\dot{\boldsymbol{p}}_i}_{=\boldsymbol{F}_i \text{ (LMB)}} = \underbrace{\sum_i \boldsymbol{v}_{P_i} \times \boldsymbol{p}_i}_{=\boldsymbol{0}} - \boldsymbol{v}_B \times \underbrace{\sum_i \boldsymbol{p}_i}_{=\boldsymbol{P}} + \underbrace{\sum_i \boldsymbol{r}_{BP_i} \times \boldsymbol{F}_i}_{=\boldsymbol{M}_B}.$$

Internal forces are equal, opposite and collinear, so their torques cancel in pairs and $\boldsymbol{M}_B$ is the net *external* torque.

Key result: AMB for a system of particles

$$\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}, \qquad \boldsymbol{P} = M\boldsymbol{v}_{\mathrm{cm}}. \tag{3.7}$$

Example 2.5: Particles on a disk (2D)

n particles m_i are fixed on a massless disk rotating about the fixed centre B with angle $\varphi(t)$; particle i sits at radius r_i. Each moves on a circle: $\boldsymbol{v}_i = r_i \dot{\varphi}\, \boldsymbol{e}_{\varphi,i}$. Then

$$\boldsymbol{H}_B = \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times \boldsymbol{p}_i = \sum_{i=1}^{n} r_i \boldsymbol{e}_{r,i} \times m_i r_i \dot{\varphi}\, \boldsymbol{e}_{\varphi,i} = \sum_{i=1}^{n} m_i r_i^2\, \dot{\varphi}\, \boldsymbol{e}_3.$$

AMB with $\boldsymbol{v}_B = \boldsymbol{0}$:

$$\boldsymbol{M}_B^{\mathrm{ext}} = M_B \boldsymbol{e}_3 = \dot{\boldsymbol{H}}_B \implies M_B^{\mathrm{ext}} = \underbrace{\Big(\sum_{i=1}^{n} m_i r_i^2\Big)}_{=I_B} \ddot{\varphi}.$$

External torque due to gravity:

$$\boldsymbol{M}_B^{\text{ext}} = \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times m_i \boldsymbol{g} = \Big(\sum_{i=1}^{n} m_i \boldsymbol{r}_{BP_i} \Big) \times \boldsymbol{g} = M \boldsymbol{r}_{B,\text{cm}} \times \boldsymbol{g}.$$

Gravity acts as if the whole mass sat at the centre of mass (this only works for a homogeneous gravity field).

Example 2.6: A ring of particles

n particles of mass $m_i = M/n$ evenly spaced on a circle of radius R:

$$I_B = \sum_{i=1}^{n} m_i R^2 = \sum_{i=1}^{n} \frac{M}{n} R^2 = MR^2.$$

As $n \to \infty$ this is a rotating thin ring.

Lecture 7 (07.10.2026)

Example 2.8: Atwood machine with driving torque

Two masses m_1 (left) and m_2 (right) hang on a rope over a massless wheel of radius R with fixed bearing B; the wheel is driven by a torque M_T. Find $\ddot{x}$ and the bearing force.
Kinematics. With x_1 measured downward on the left and x_2 upward on the right: $x_1 = R\varphi$, $x_2 = R\varphi \Rightarrow x_1 = x_2 =: x$. 1 DOF.
Statics (for comparison): $B_2 = (m_1 + m_2)g$, $B_1 = 0$ – *not valid in dynamics.*
Dynamics. Cut the rope; the rope forces are S_1 (left) and S_2 (right).

$$\begin{aligned}
\text{LMB } m_1: &\quad m_1\ddot{x} = m_1 g - S_1, \\
\text{LMB } m_2: &\quad m_2\ddot{x} = -m_2 g + S_2, \\
\text{LMB wheel:} &\quad B_2 = S_1 + S_2 \quad (\text{wheel is massless} \Rightarrow \text{force equilibrium}), \\
\text{AMB wheel about } B: &\quad \underbrace{I_B}_{=0}\ddot{\varphi} = S_1 R - S_2 R + M_T = 0 \;\Rightarrow\; S_1 - S_2 = -\frac{M_T}{R}.
\end{aligned}$$

Add the two particle equations: $(m_1 + m_2)\ddot{x} = (m_1 - m_2)g - S_1 + S_2 = (m_1 - m_2)g + M_T/R$, so

$$\boxed{\ddot{x} = \frac{m_1 - m_2}{m_1 + m_2} g + \frac{M_T}{(m_1 + m_2)R}}. \qquad (*)$$

Bearing force:

$$B_2 = S_1 + S_2 = m_1 g - m_1\ddot{x} + m_2\ddot{x} + m_2 g = \underbrace{(m_1 + m_2)g}_{\text{statics}} + \underbrace{(m_2 - m_1)\ddot{x}}_{\text{because it's moving}},$$

with $\ddot{x}$ from $(*)$.
Alternative: WEB. The work done by a torque: a torque M_T is equivalent to a force couple F at distance R on both sides ($M_T = 2FR$). Rotating by $\mathrm{d}\varphi$, each force moves $\mathrm{d}r = R\,\mathrm{d}\varphi$, so $2F\,\mathrm{d}r = 2FR\,\mathrm{d}\varphi = M_T\,\mathrm{d}\varphi$. Hence

$$T(t) + V(t) - [T(t_0) + V(t_0)] = \int_{\varphi_0}^{\varphi} M_T\,\mathrm{d}\varphi = M_T(\varphi - \varphi_0):$$

$$\frac{m_1}{2}\dot{x}^2 + \frac{m_2}{2}\dot{x}^2 + m_2 g x - m_1 g x - [\text{initial const.}] = M_T(\varphi - \varphi_0).$$

Differentiate with respect to time ($\dot\varphi = \dot x/R$):

$$m_1\dot x\ddot x + m_2\dot x\ddot x + m_2 g\dot x - m_1 g\dot x = M_T\frac{\dot x}{R} \implies (m_1+m_2)\ddot x = (m_1-m_2)g + \frac{M_T}{R}. \quad \checkmark$$

Treat as one system. With $\boldsymbol{e}_2$ up and the wheel centre as origin, the masses hang at x-positions $-R$ and $+R$. LMB:

$$(m_1+m_2)\boldsymbol{a}_{\text{cm}} = B_1\boldsymbol{e}_1 + (B_2 - m_1 g - m_2 g)\boldsymbol{e}_2.$$

The vertical positions are $-x$ (left, going down) and $+x$ (right, going up) relative to a reference, so

$$\boldsymbol{r}_{\text{cm}} = \frac{-m_1R + m_2R}{m_1+m_2}\boldsymbol{e}_1 + \frac{m_2x - m_1x}{m_1+m_2}\boldsymbol{e}_2 + \text{const} \implies \boldsymbol{a}_{\text{cm}} = \frac{m_2-m_1}{m_1+m_2}\ddot x\,\boldsymbol{e}_2.$$

Hence $(m_2-m_1)\ddot x = B_2 - m_1 g - m_2 g$ and $B_1 = 0$. ✓ (same as before)

3.3 Particle collisions

Two smooth spheres collide (small velocities, no friction). The collision plane is tangent to both at the contact point, $\boldsymbol{e}_n$ is the common normal and $\boldsymbol{e}_t$ lies in the plane. Particle 2 receives the impulse $\hat F\boldsymbol{e}_n$, particle 1 receives $-\hat F\boldsymbol{e}_n$ (Newton 3):

$$m_1\boldsymbol{v}_1(t_+) - m_1\boldsymbol{v}_1(t_-) = -\hat F\boldsymbol{e}_n, \qquad m_2\boldsymbol{v}_2(t_+) - m_2\boldsymbol{v}_2(t_-) = \hat F\boldsymbol{e}_n.$$

Tangential components are unchanged:

$$\boxed{v_1^t(t_+) = v_1^t(t_-), \qquad v_2^t(t_+) = v_2^t(t_-)}. \tag{3.8}$$

Normal components, split into compression and restitution:

$$\begin{aligned}
\text{compression:}\quad & m_1v_1^n(t_0) - m_1v_1^n(t_-) = -\hat F_{\text{comp}}, \qquad m_2v_2^n(t_0) - m_2v_2^n(t_-) = \hat F_{\text{comp}},\\
\text{restitution:}\quad & m_1v_1^n(t_+) - m_1v_1^n(t_0) = -\hat F_{\text{rest}}, \qquad m_2v_2^n(t_+) - m_2v_2^n(t_0) = \hat F_{\text{rest}}.
\end{aligned}$$

4 equations but 6 unknowns (everything except v_1^n, v_2^n at t_-). Two further conditions:

1. material law: $e = \hat F_{\text{rest}}/\hat F_{\text{comp}}$;
2. at maximum compression the two bodies move together: $v_1^n(t_0) = v_2^n(t_0) =: v_0$.

Solving. Adding the two compression equations eliminates $\hat F_{\text{comp}}$: $(m_1+m_2)v_0 = m_1v_1^n(t_-) + m_2v_2^n(t_-)$, i.e. v_0 is the cm velocity. Then $\hat F_{\text{comp}} = m_2(v_0 - v_2^n(t_-)) = \frac{m_1m_2}{m_1+m_2}(v_1^n(t_-) - v_2^n(t_-))$ and $\hat F_{\text{rest}} = e\hat F_{\text{comp}}$. With $v_1^n(t_+) = v_1^n(t_-) - (1+e)\hat F_{\text{comp}}/m_1$ etc.:

Key result: Normal velocities after collision

$$v_1^n(t_+) = \frac{m_1 - e\,m_2}{m_1+m_2}\,v_1^n(t_-) + \frac{(1+e)\,m_2}{m_1+m_2}\,v_2^n(t_-), \tag{3.9}$$

$$v_2^n(t_+) = \frac{m_2 - e\,m_1}{m_1+m_2}\,v_2^n(t_-) + \frac{(1+e)\,m_1}{m_1+m_2}\,v_1^n(t_-), \tag{3.10}$$

and, equivalently (subtract the two equations),

$$e = -\frac{v_2^n(t_+) - v_1^n(t_+)}{v_2^n(t_-) - v_1^n(t_-)} \qquad \text{(ratio of relative normal velocities after/before).} \tag{3.11}$$

Also note: there are no external forces in the normal direction, so linear momentum is conserved:

$$m_1 v_1^n(t_+) + m_2 v_2^n(t_+) = m_1 v_1^n(t_-) + m_2 v_2^n(t_-). \tag{3.12}$$

Special cases (Examples 2.9, 2.10).

- **Rigid wall:** $v_2^n = 0$, $m_2 \to \infty$: $v_1^n(t_+) = -e\, v_1^n(t_-)$ (as in Section 1).
- **Plastic collision** ($e = 0$): both move together, $v_1^n(t_+) = v_2^n(t_+) = \dfrac{m_1 v_1^n(t_-) + m_2 v_2^n(t_-)}{m_1 + m_2}$.
- **Elastic collision with equal masses** ($e = 1$, $m_1 = m_2$): they switch velocities, $v_1^n(t_+) = v_2^n(t_-)$, $v_2^n(t_+) = v_1^n(t_-)$ (Newton's cradle).

3.4 Particles with variable mass

Mass accretion. A body of mass $m(t)$ and velocity $\boldsymbol{v}(t)$ absorbs in $\mathrm{d}t$ a mass $\mathrm{d}m$ that arrives with velocity $\boldsymbol{v}_m(t)$. Apply the LMB to the closed system "body + $\mathrm{d}m$":

$$\begin{aligned} \boldsymbol{p}(t) &= m(t)\boldsymbol{v}(t) + \mathrm{d}m\, \boldsymbol{v}_m(t), \\ \boldsymbol{p}(t+\mathrm{d}t) &= m(t+\mathrm{d}t)\boldsymbol{v}(t+\mathrm{d}t) = [m + \dot{m}\,\mathrm{d}t + O(\mathrm{d}t^2)]\,[\boldsymbol{v} + \dot{\boldsymbol{v}}\,\mathrm{d}t + O(\mathrm{d}t^2)] \quad \text{(Taylor)} \\ &= m\boldsymbol{v} + \dot{m}\boldsymbol{v}\,\mathrm{d}t + m\dot{\boldsymbol{v}}\,\mathrm{d}t + O(\mathrm{d}t^2). \end{aligned}$$

Hence, with $\mathrm{d}m = \dot{m}\,\mathrm{d}t$,

$$\dot{\boldsymbol{p}} = \lim_{\mathrm{d}t\to 0} \frac{\boldsymbol{p}(t+\mathrm{d}t) - \boldsymbol{p}(t)}{\mathrm{d}t} = \dot{m}\boldsymbol{v} + m\dot{\boldsymbol{v}} - \dot{m}\boldsymbol{v}_m.$$

Key result: LMB with variable mass / rocket equation

$$\sum_i \boldsymbol{F}_i = m\boldsymbol{a} + \dot{m}(\boldsymbol{v} - \boldsymbol{v}_m) = m\boldsymbol{a} - \dot{m}\,\boldsymbol{v}_{\mathrm{rel}}, \qquad \boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v}_m - \boldsymbol{v}, \tag{3.13}$$

$$\iff \quad m\boldsymbol{a} = \sum_i \boldsymbol{F}_i + \underbrace{\dot{m}\,\boldsymbol{v}_{\mathrm{rel}}}_{=\text{thrust}}. \tag{3.14}$$

Common pitfall

$\sum \boldsymbol{F} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v})$ is **wrong** for variable mass: it is not Galilean invariant. Always use the relative velocity of the gained/lost mass.

Example 2.11: Rocket launch

A rocket of mass $m(t)$ moves up (x up, $\boldsymbol{v} = \dot{x}\boldsymbol{e}_2$) and ejects gas downward relative to the rocket: $\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v}_{\mathrm{gas}} - \boldsymbol{v}_{\mathrm{rocket}} = -c\,\boldsymbol{e}_2$, $c > 0$. External forces: gravity $-mg\boldsymbol{e}_2$ and drag $-F_{\mathrm{drag}}\boldsymbol{e}_2$.

$$\sum \boldsymbol{F}_i = (-F_{\mathrm{drag}} - mg)\boldsymbol{e}_2 \overset{!}{=} m\ddot{x}\,\boldsymbol{e}_2 + \dot{m}c\,\boldsymbol{e}_2 \implies \boxed{m\ddot{x} = -F_{\mathrm{drag}} - mg - \dot{m}c}.$$

Because the rocket loses mass, $\dot{m} < 0$, so $-\dot{m}c > 0$ is the thrust; it must exceed $mg + F_{\mathrm{drag}}$ to accelerate the rocket upward.

4 Dynamics of Rigid Bodies

Lecture 8 (12.10.2026)

A rigid body is the limit of a system of particles ($n \to \infty$, $\Delta m_i \to 0$) in which all distances are fixed: $|\boldsymbol{r}_{AB}| = \text{const}$ for all points A, B of the body $\mathcal{B}$.

4.1 Kinematics

Definition: Total mass and centre of mass of a body

With density $\rho = \mathrm{d}m/\mathrm{d}V$:

$$M = \lim_{\substack{n\to\infty \\ \Delta m_i \to 0}} \sum_{i=1}^{n} \Delta m_i = \int_{\mathcal{B}} \mathrm{d}m = \int_{\mathcal{B}} \rho \, \mathrm{d}V, \qquad \boldsymbol{r}_{\mathrm{cm}}(t) = \frac{1}{M} \int_{\mathcal{B}} \boldsymbol{r}(t) \, \mathrm{d}m. \tag{4.1}$$

Key result: DOFs of rigid bodies

An unconstrained rigid body in d dimensions has $\dfrac{d(d+1)}{2}$ DOFs: d translations plus $\frac{d(d-1)}{2}$ rotations.

- 1D: 1 DOF, 2D: 3 DOFs (x_1, x_2, φ), 3D: 6 DOFs (3 translations, 3 rotation angles).

n rigid bodies in d dimensions with k independent holonomic constraints have

$$\mathrm{DOF} = n\,\frac{d(d+1)}{2} - k. \tag{4.2}$$

How are the motions of points on a rigid body related? For two points A, B of the body:

$$\boldsymbol{r}_B = \boldsymbol{r}_A + \boldsymbol{r}_{AB}, \qquad \boldsymbol{v}_B = \boldsymbol{v}_A + \dot{\boldsymbol{r}}_{AB}, \qquad \boldsymbol{a}_B = \boldsymbol{a}_A + \ddot{\boldsymbol{r}}_{AB}.$$

Since $|\boldsymbol{r}_{AB}|$ is constant, $\boldsymbol{r}_{AB}$ can only change its direction – $\dot{\boldsymbol{r}}_{AB}$ must be related to rotation (without rotation these terms are zero).

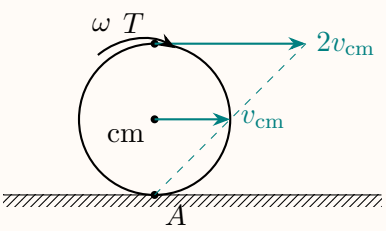
Rotations (recall Mechanics I). Define the angular velocity vector $\boldsymbol{\omega}(t) = \omega(t)\,\boldsymbol{e}_\omega(t)$ ($\boldsymbol{e}_\omega$: rotation axis; in 2D, $\boldsymbol{e}_\omega = \boldsymbol{e}_3$ is constant, out of the plane). For a body rotating about a fixed point Π,

$$\boldsymbol{v}_B = \boldsymbol{\omega} \times \boldsymbol{r}_{\Pi B}, \qquad \boldsymbol{v}_A = \boldsymbol{\omega} \times \boldsymbol{r}_{\Pi A} \implies \boldsymbol{v}_B - \boldsymbol{v}_A = \boldsymbol{\omega} \times \underbrace{(\boldsymbol{r}_{\Pi B} - \boldsymbol{r}_{\Pi A})}_{=\boldsymbol{r}_{AB}}.$$

Key result: Velocity transfer formula

$$\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{AB} \qquad \text{(valid in 2D and 3D, for any two points of the body).} \tag{4.3}$$

Example 3.1: Rolling disk (rolling without slipping)



Kinematics: $x_{\text{cm}} = R\varphi$ (φ: rotation angle), so $\dot{x}_{\text{cm}} = R\dot{\varphi} = R\omega$. With $\boldsymbol{\omega} = -\omega \boldsymbol{e}_3$ (clockwise when rolling in $+\boldsymbol{e}_1$) and $\boldsymbol{v}_{\text{cm}} = R\omega \boldsymbol{e}_1$:

$$\boldsymbol{v}_T = \boldsymbol{v}_{\text{cm}} + \boldsymbol{\omega} \times \boldsymbol{r}_{\text{cm},T} = R\omega \boldsymbol{e}_1 + (-\omega \boldsymbol{e}_3) \times R\boldsymbol{e}_2 = R\omega \boldsymbol{e}_1 + R\omega \boldsymbol{e}_1 = 2R\omega \boldsymbol{e}_1,$$
$$\boldsymbol{v}_A = \boldsymbol{v}_{\text{cm}} + \boldsymbol{\omega} \times \boldsymbol{r}_{\text{cm},A} = R\omega \boldsymbol{e}_1 + (-\omega \boldsymbol{e}_3) \times (-R\boldsymbol{e}_2) = R\omega \boldsymbol{e}_1 - R\omega \boldsymbol{e}_1 = \boldsymbol{0}.$$

The contact point is momentarily at rest; the velocity profile along the vertical diameter is linear.

Key result: Instantaneous centre of rotation (ICR)

For every rigid body in 2D (with $\omega \neq 0$) there exists a point Π with $\boldsymbol{v}_\Pi = \boldsymbol{0}$, so that $\boldsymbol{v}_P = \boldsymbol{\omega} \times \boldsymbol{r}_{\Pi P}$ for every point P. Given one point A with velocity $\boldsymbol{v}_A$:

$$\boldsymbol{r}_{A\Pi} = \frac{1}{\omega}(\boldsymbol{e}_\omega \times \boldsymbol{v}_A). \tag{4.4}$$

(The ICR lies on the line through A perpendicular to $\boldsymbol{v}_A$, at distance $|\boldsymbol{v}_A|/\omega$.)

Check: $\boldsymbol{v}_\Pi = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{A\Pi} = \boldsymbol{v}_A + \boldsymbol{e}_\omega \times (\boldsymbol{e}_\omega \times \boldsymbol{v}_A) = \boldsymbol{v}_A - \boldsymbol{v}_A = \boldsymbol{0}$, using $\boldsymbol{e}_\omega \times (\boldsymbol{e}_\omega \times \boldsymbol{v}) = -\boldsymbol{v}$ for $\boldsymbol{v} \perp \boldsymbol{e}_\omega$.

Key result: Theorem of projected velocities

For a rigid body, $\boldsymbol{r}_{AB} \cdot \boldsymbol{r}_{AB} = \text{const}$. Differentiating:

$$2\,\dot{\boldsymbol{r}}_{AB} \cdot \boldsymbol{r}_{AB} = 0 \iff (\boldsymbol{v}_B - \boldsymbol{v}_A) \cdot \boldsymbol{r}_{AB} = 0 \iff \boxed{\boldsymbol{v}_B \cdot \boldsymbol{r}_{AB} = \boldsymbol{v}_A \cdot \boldsymbol{r}_{AB}}.$$

The velocity components of two points along their connecting line are equal (Mechanics I).

Example 3.2: Slider–crank mechanism

A crank AB (length L) rotates about the fixed pivot A with angular velocity $\boldsymbol{\omega}_2$; it makes the angle α with the ground. The connecting rod BC (length $2L$, angle β to the horizontal) drives a slider C moving horizontally. Given ω_2, find the slider speed v_C ($\boldsymbol{v}_C = -v_C \boldsymbol{e}_1$).

Geometry. With $\boldsymbol{e}_1$ to the right and $\boldsymbol{e}_2$ up (B is above-left of A, C is below-left of B):

$$\boldsymbol{r}_{AB} = -L\cos\alpha\,\boldsymbol{e}_1 + L\sin\alpha\,\boldsymbol{e}_2, \qquad \boldsymbol{r}_{BC} = -2L\cos\beta\,\boldsymbol{e}_1 - 2L\sin\beta\,\boldsymbol{e}_2,$$

and since C stays at the height of A: $L\sin\alpha = 2L\sin\beta$.

Velocity of B (A fixed, $\boldsymbol{\omega}_2 = \omega_2 \boldsymbol{e}_3$):

$$\boldsymbol{v}_B = \boldsymbol{\omega}_2 \times \boldsymbol{r}_{AB} = -L\omega_2(\sin\alpha\,\boldsymbol{e}_1 + \cos\alpha\,\boldsymbol{e}_2).$$

Method 1 – projected velocities along BC: $\boldsymbol{v}_B \cdot \boldsymbol{r}_{BC} = \boldsymbol{v}_C \cdot \boldsymbol{r}_{BC}$:

$$\begin{pmatrix} -L\omega_2\sin\alpha \\ -L\omega_2\cos\alpha \end{pmatrix} \cdot \begin{pmatrix} -2L\cos\beta \\ -2L\sin\beta \end{pmatrix} = \begin{pmatrix} -v_C \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -2L\cos\beta \\ -2L\sin\beta \end{pmatrix}$$

$$\Longrightarrow\ 2L^2\omega_2(\sin\alpha\cos\beta + \cos\alpha\sin\beta) = 2Lv_C\cos\beta$$

$$\Longrightarrow\ \boxed{v_C = \omega_2 L(\sin\alpha + \cos\alpha\tan\beta) = \omega_2 L\frac{\sin(\alpha+\beta)}{\cos\beta}}.$$

Method 2 – velocity transfer formula (gives also the rod's angular velocity $\boldsymbol{\omega}_1$):

$$\boldsymbol{v}_C = \boldsymbol{v}_B + \boldsymbol{\omega}_1 \times \boldsymbol{r}_{BC} = \underbrace{\boldsymbol{v}_A}_{=\boldsymbol{0}} + \boldsymbol{\omega}_2 \times \boldsymbol{r}_{AB} + \boldsymbol{\omega}_1 \times \boldsymbol{r}_{BC} \overset{!}{=} -v_C \boldsymbol{e}_1,$$

two scalar equations for the two unknowns v_C and ω_1. Note: $\boldsymbol{\omega}_1 = \dot{\beta}\,\boldsymbol{e}_3$, $\boldsymbol{\omega}_2 = -\dot{\alpha}\,\boldsymbol{e}_3$.

Acceleration transfer formula. Differentiate $\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{AB}$ in time, using $\dot{\boldsymbol{r}}_{AB} = \boldsymbol{\omega} \times \boldsymbol{r}_{AB}$:

Key result: Acceleration transfer formula

$$\boldsymbol{a}_B = \boldsymbol{a}_A + \dot{\boldsymbol{\omega}} \times \boldsymbol{r}_{AB} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}_{AB}). \tag{4.5}$$

In 2D ($\boldsymbol{\omega} \perp \boldsymbol{r}_{AB}$): $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}_{AB}) = -\omega^2 \boldsymbol{r}_{AB}$ (centripetal term).

Example 3.3: Rolling disk (accelerations)

$\boldsymbol{v}_{\text{cm}} = R\omega \boldsymbol{e}_1$, $\boldsymbol{a}_{\text{cm}} = R\dot{\omega} \boldsymbol{e}_1$, $\boldsymbol{\omega} = -\omega \boldsymbol{e}_3$, $\dot{\boldsymbol{\omega}} = -\dot{\omega} \boldsymbol{e}_3$.

$$\boldsymbol{a}_T = \boldsymbol{a}_{\text{cm}} + \dot{\boldsymbol{\omega}} \times \boldsymbol{r}_{\text{cm},T} - \omega^2 \boldsymbol{r}_{\text{cm},T} = R\dot{\omega} \boldsymbol{e}_1 + (-\dot{\omega} \boldsymbol{e}_3) \times R\boldsymbol{e}_2 - \omega^2 R\boldsymbol{e}_2 = 2R\dot{\omega} \boldsymbol{e}_1 - R\omega^2 \boldsymbol{e}_2,$$

$$\boldsymbol{a}_A = R\dot{\omega} \boldsymbol{e}_1 + (-\dot{\omega} \boldsymbol{e}_3) \times (-R\boldsymbol{e}_2) - \omega^2 (-R\boldsymbol{e}_2) = \boxed{R\omega^2 \boldsymbol{e}_2}.$$

The contact point has *zero velocity but non-zero acceleration* – it is accelerated upward, towards the centre. (The top point T must accelerate inward too, because its velocity rotates.)

Common pitfall

The ICR has $\boldsymbol{v}_\Pi = \boldsymbol{0}$ but in general $\boldsymbol{a}_\Pi \neq \boldsymbol{0}$. Do not use $\boldsymbol{a} = \boldsymbol{0}$ at the contact point of a rolling wheel.

Lecture 9 (14.10.2026)

Example 3.5: Ladder sliding down a wall

Two blocks of mass m (A on a frictionless vertical wall, B on the frictionless floor) are connected by a massless rigid rod of length L; φ is the angle of the rod from the vertical. Released from (almost) vertical, $\varphi \approx 0$, at rest. Find v_A, v_B, a_A, a_B and the forces.
DOF: 2 particles $\times$ 2D $= 4$, minus 3 constraints (A on wall, B on floor, rod) $\Rightarrow$ 1 DOF: $\varphi(t)$.
Velocities. $\boldsymbol{r}_A = L\cos\varphi\, \boldsymbol{e}_2$, $\boldsymbol{r}_B = L\sin\varphi\, \boldsymbol{e}_1$:

$$\boldsymbol{v}_A = -\dot{\varphi} L \sin\varphi\, \boldsymbol{e}_2, \qquad \boldsymbol{v}_B = \dot{\varphi} L \cos\varphi\, \boldsymbol{e}_1, \qquad \boldsymbol{\omega} = \dot{\varphi}\, \boldsymbol{e}_3 \text{ (up to sign convention)}.$$

WEB (conservative; zero potential at floor level, only A changes height):

$$mgL = mgL\cos\varphi + \frac{m}{2} v_A^2 + \frac{m}{2} v_B^2 = mgL\cos\varphi + \frac{m}{2} \dot{\varphi}^2 L^2 (\sin^2\varphi + \cos^2\varphi)$$

$$\Longrightarrow \frac{m}{2} \dot{\varphi}^2 L^2 = mgL(1 - \cos\varphi) \Longrightarrow \boxed{\dot{\varphi}^2 = \frac{2g}{L}(1 - \cos\varphi)}.$$

Differentiating: $2\dot{\varphi}\ddot{\varphi} = \frac{2g}{L} \sin\varphi\, \dot{\varphi} \Rightarrow \dot{\omega} = \ddot{\varphi} = \frac{g}{L} \sin\varphi$.
Accelerations. Differentiate the velocities (equivalent to the transfer formula with $\boldsymbol{r}_{AB} = L\sin\varphi\, \boldsymbol{e}_1 - L\cos\varphi\, \boldsymbol{e}_2$):

$$\begin{aligned} a_A &:= -\ddot{x}_{2,A} = \ddot{\varphi} L \sin\varphi + \dot{\varphi}^2 L \cos\varphi \quad \text{(downward)} \\ &= g\sin^2\varphi + 2g(1 - \cos\varphi)\cos\varphi = \boxed{g(1 + 2\cos\varphi - 3\cos^2\varphi)}, \\ a_B &:= \ddot{x}_{1,B} = \ddot{\varphi} L \cos\varphi - \dot{\varphi}^2 L \sin\varphi \\ &= g\sin\varphi\cos\varphi - 2g(1 - \cos\varphi)\sin\varphi = \boxed{g\sin\varphi(3\cos\varphi - 2)}. \end{aligned}$$

Forces. Let S be the rod force (tension positive), H the wall force on A, V the floor force on B. LMB on A (vertical) and on B (horizontal; V does not appear there):

$$ma_A = mg + S\cos\varphi, \qquad ma_B = -S\sin\varphi \Longrightarrow \boxed{S = mg(2 - 3\cos\varphi)}.$$

Plugging in: $V = mg - S\cos\varphi$, $H = -S\sin\varphi$.
Interpretation. For $\cos\varphi > 2/3$, $S < 0$: the rod is in compression and pushes A against the wall. At $\cos\varphi = 2/3$ the wall force H becomes zero; beyond that it would have to pull, which a wall cannot do – block A leaves the wall at $\varphi = \arccos(2/3) \approx 48.2°$ and the above solution is no longer valid.

Rigid body kinematics in 3D

The velocity transfer formula $\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{AB}$ holds in 3D, but now the axis can change: $\boldsymbol{\omega}(t) = \omega(t)\,\boldsymbol{e}_\omega(t)$ with $|\boldsymbol{e}_\omega| = 1$ defining the *instantaneous axis of rotation* (IAR).

Key result

Unlike the ICR in 2D (which is stationary, $\boldsymbol{v} = \boldsymbol{0}$), points on the IAR can have a velocity – but only along the axis:

$$\boldsymbol{v}_{\text{IAR}} = v_{\text{IAR}}\,\boldsymbol{e}_\omega \tag{4.6}$$

(screw motion). In 3D, if two points of a body have zero velocity, the IAR passes through both.

Example 3.6: Rolling cone

A cone with apex O fixed at height R on a vertical axis has its axis horizontal (length L) and its base circle (radius R) touching the ground at A; it rolls around the vertical axis with precession angle θ and spins about its own axis with angle φ. Given $\dot\theta$, find $\boldsymbol{\omega}$.
For any point C of the cone: $\boldsymbol{v}_C = \underbrace{\boldsymbol{v}_O}_{=\boldsymbol{0}} + \boldsymbol{\omega} \times \boldsymbol{r}_{OC}$. Rolling: the contact point A is momentarily at rest, so

$$\boldsymbol{v}_A = \boldsymbol{\omega} \times \boldsymbol{r}_{OA} = \boldsymbol{0} \implies \boldsymbol{\omega} \parallel \boldsymbol{r}_{OA} \quad (\text{both} \neq \boldsymbol{0}).$$

The ray through O and A *must be the IAR*: two points with zero velocity on the same body.
Finding $\boldsymbol{\omega}$. The rotation is the sum of the precession about the vertical and the spin about the cone axis $\boldsymbol{e}_r$:

$$\boldsymbol{\omega} = \dot\theta\,\boldsymbol{e}_3 - \dot\varphi\,\boldsymbol{e}_r.$$

Since $\boldsymbol{r}_{OA} = L\boldsymbol{e}_r - R\boldsymbol{e}_3$, parallelism requires $\dot\theta/\dot\varphi = R/L = \tan\alpha$ (α: half-angle of the cone), i.e. $R\dot\varphi = L\dot\theta$ (rolling condition). Therefore

$$\boxed{\boldsymbol{\omega} = \dot\theta\,\boldsymbol{e}_3 - \frac{L}{R}\dot\theta\,\boldsymbol{e}_r}.$$

Angular velocities are additive (as vectors).

4.2 Kinetics of rigid bodies

4.2.1 Balance of linear momentum

The total linear momentum of a rigid body $\mathcal{B}$:

$$\boldsymbol{P} = \lim_{\substack{n\to\infty \\ \Delta m_i \to 0}} \sum_{i=1}^{n} \boldsymbol{v}_i \Delta m_i = \int_{\mathcal{B}} \boldsymbol{v}\,\mathrm{d}m = \int_{\mathcal{B}} \boldsymbol{v}\rho\,\mathrm{d}V = M\boldsymbol{v}_{\text{cm}},$$

using $\boldsymbol{r}_{\text{cm}} = \frac{1}{M}\int_{\mathcal{B}} \boldsymbol{r}\,\mathrm{d}m$ and differentiating.

Key result: LMB for a rigid body

$$\boldsymbol{P} = M\boldsymbol{v}_{\mathrm{cm}}, \qquad \frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} = \sum_i \boldsymbol{F}_i^{\mathrm{ext}} \overset{M=\text{const}}{\Longrightarrow} M\boldsymbol{a}_{\mathrm{cm}} = \sum_i \boldsymbol{F}_i^{\mathrm{ext}}. \tag{4.7}$$

The body behaves like a "super particle" at its centre of mass. Example: only gravity $\Rightarrow$ $M\boldsymbol{a}_{\mathrm{cm}} = M\boldsymbol{g} \Rightarrow \boldsymbol{a}_{\mathrm{cm}} = \boldsymbol{g}$.

4.2.2 Balance of angular momentum

Lecture 10 (19.10.2026)

Recall the AMB for a system of particles, $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$ with $\boldsymbol{P} = M\boldsymbol{v}_{\mathrm{cm}}$ ($\boldsymbol{M}_B$: external torque). The angular momentum of a rigid body about a point B *of the body*:

$$\boldsymbol{H}_B = \lim \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times \Delta m_i \boldsymbol{v}_i = \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B) \times \boldsymbol{v}\, \mathrm{d}m.$$

Insert the velocity transfer formula $\boldsymbol{v} = \boldsymbol{v}_B + \boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_B)$:

$$\boldsymbol{H}_B = \underbrace{\int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B)\, \mathrm{d}m \times \boldsymbol{v}_B}_{(1)} + \underbrace{\int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B) \times \left[\boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_B)\right] \mathrm{d}m}_{(2)}.$$

Term (1): $\left[\int_{\mathcal{B}} \boldsymbol{r}\, \mathrm{d}m - \int_{\mathcal{B}} \boldsymbol{r}_B\, \mathrm{d}m\right] \times \boldsymbol{v}_B = M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B$.

Term (2): Let $\bar{\boldsymbol{r}} = \boldsymbol{r} - \boldsymbol{r}_B$. Using the identity $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$:

$$\bar{\boldsymbol{r}} \times (\boldsymbol{\omega} \times \bar{\boldsymbol{r}}) = (\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}})\boldsymbol{\omega} - \bar{\boldsymbol{r}}(\bar{\boldsymbol{r}} \cdot \boldsymbol{\omega}) = \left[(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}})\underline{\underline{\mathbb{1}}} - \bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}}\right]\boldsymbol{\omega},$$

where $\underline{\underline{\mathbb{1}}}$ is the unit tensor (identity matrix) and $\bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}}$ the dyadic (outer) product. Since $\boldsymbol{\omega}$ does not depend on the position in the body, it can be pulled out of the integral:

Definition: Mass moment of inertia tensor

$$\underline{\underline{I}}_B = \int_{\mathcal{B}} \left[(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}})\underline{\underline{\mathbb{1}}} - \bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}}\right] \mathrm{d}m, \qquad \bar{\boldsymbol{r}} = \boldsymbol{r} - \boldsymbol{r}_B. \tag{4.8}$$

Key result: Angular momentum and AMB of a rigid body

$$\boldsymbol{H}_B = M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B + \underline{\underline{I}}_B\, \boldsymbol{\omega}, \tag{4.9}$$

$$\boldsymbol{M}_B = \frac{\mathrm{d}}{\mathrm{d}t}\left[M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B + \underline{\underline{I}}_B\, \boldsymbol{\omega}\right] + \boldsymbol{v}_B \times \boldsymbol{P}. \tag{4.10}$$

Components of $\underline{\underline{I}}_B$. With $\bar{\boldsymbol{r}} = (\bar{x}_1, \bar{x}_2, \bar{x}_3)$, $(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}}) = \bar{x}_1^2 + \bar{x}_2^2 + \bar{x}_3^2$ and $(\bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}})_{ij} = \bar{x}_i \bar{x}_j$:

$$[\underline{\underline{I}}_B] = \int_{\mathcal{B}} \begin{bmatrix} \bar{x}_2^2 + \bar{x}_3^2 & -\bar{x}_1\bar{x}_2 & -\bar{x}_1\bar{x}_3 \\ -\bar{x}_1\bar{x}_2 & \bar{x}_1^2 + \bar{x}_3^2 & -\bar{x}_2\bar{x}_3 \\ -\bar{x}_1\bar{x}_3 & -\bar{x}_2\bar{x}_3 & \bar{x}_1^2 + \bar{x}_2^2 \end{bmatrix} \rho\, \mathrm{d}V. \tag{4.11}$$

Diagonal entries: moments of inertia about the axes (sum of squared distances *to the axis*); off-diagonal: products of inertia (deviation moments).

Special cases.

- B = centre of mass: $\boldsymbol{r}_{\mathrm{cm}} = \boldsymbol{r}_B$, the first term vanishes, and $\boldsymbol{v}_B \times \boldsymbol{P} = \boldsymbol{v}_{\mathrm{cm}} \times M\boldsymbol{v}_{\mathrm{cm}} = \boldsymbol{0}$:

$$\boxed{\boldsymbol{M}_{\mathrm{cm}} = \frac{\mathrm{d}}{\mathrm{d}t}(\underline{\underline{I}}_{\mathrm{cm}}\boldsymbol{\omega})} \tag{4.12}$$

- B stationary, $\boldsymbol{v}_B = \boldsymbol{0}$: same form, $\boldsymbol{M}_B = \frac{\mathrm{d}}{\mathrm{d}t}(\underline{\underline{I}}_B\boldsymbol{\omega})$. *Always choose B like that (one of these two)!*
- If additionally $\underline{\underline{I}}_B = \text{const}$: $\boldsymbol{M}_B = \underline{\underline{I}}_B\dot{\boldsymbol{\omega}}$. (Recall 2D: $M_B = I_B\dot{\omega}$.)
- If $B = \mathrm{cm}$ or $\boldsymbol{v}_B = \boldsymbol{0}$, and no torque: $\underline{\underline{I}}_B\boldsymbol{\omega} = \text{const}$ (angular momentum conserved).

Why is $\underline{\underline{I}}_B$ a tensor? If the axes are chosen such that $\underline{\underline{I}}_{\mathrm{cm}}$ is diagonal, the 3D AMB decouples:

$$\begin{pmatrix} M_{\mathrm{cm},1} \\ M_{\mathrm{cm},2} \\ M_{\mathrm{cm},3} \end{pmatrix} = \begin{bmatrix} I_{\mathrm{cm},1} & & \\ & I_{\mathrm{cm},2} & \\ & & I_{\mathrm{cm},3} \end{bmatrix} \begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{pmatrix} \quad (\text{if } \underline{\underline{I}} = \text{const}).$$

In general, however, $\underline{\underline{I}}$ maps $\boldsymbol{\omega}$ to $\boldsymbol{H}$ in a *different direction* – a scalar cannot do that. The resistance to rotation depends on the axis, so a direction-dependent linear map (a 2nd-order tensor) is needed.

Example 3.7: Moment of inertia of a solid cylinder

Homogeneous cylinder, radius R, length L, axis along x_3, origin at the cm, $M = \rho\pi R^2 L$. Cylindrical coordinates: $x_1 = r\cos\varphi$, $x_2 = r\sin\varphi$, $x_3 = x_3$, $\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}x_3$.

$$[\underline{\underline{I}}_{\mathrm{cm}}] = \int_0^R \int_{-L/2}^{L/2} \int_0^{2\pi} \begin{bmatrix} r^2\sin^2\varphi + x_3^2 & -r^2\sin\varphi\cos\varphi & -r\cos\varphi\, x_3 \\ & r^2\cos^2\varphi + x_3^2 & -r\sin\varphi\, x_3 \\ \text{sym.} & & r^2 \end{bmatrix} \rho\, r\,\mathrm{d}\varphi\,\mathrm{d}x_3\,\mathrm{d}r.$$

- Off-diagonals vanish: $\int_0^{2\pi}\sin\varphi\cos\varphi\,\mathrm{d}\varphi = 0$, $\int_0^{2\pi}\cos\varphi\,\mathrm{d}\varphi = 0$ (and $\int x_3\,\mathrm{d}x_3 = 0$).
- $I_{33} = \rho\int_0^R r^3\,\mathrm{d}r \cdot 2\pi \cdot L = \rho\frac{R^4}{4}2\pi L = \underbrace{\rho\pi R^2 L}_{M}\frac{R^2}{2} = \frac{MR^2}{2}$.
- I_{11}: $\rho\int r^3\,\mathrm{d}r\int\sin^2\varphi\,\mathrm{d}\varphi\int\mathrm{d}x_3 = \rho\frac{R^4}{4}\pi L = M\frac{R^2}{4}$, and $\rho\int r\,\mathrm{d}r\int\mathrm{d}\varphi\int x_3^2\,\mathrm{d}x_3 = \rho\frac{R^2}{2}2\pi\frac{2}{3}\left(\frac{L}{2}\right)^3 = M\frac{L^2}{12}$.

$$\boxed{[\underline{\underline{I}}_{\mathrm{cm}}] = \begin{bmatrix} \frac{MR^2}{4} + \frac{ML^2}{12} & 0 & 0 \\ 0 & \frac{MR^2}{4} + \frac{ML^2}{12} & 0 \\ 0 & 0 & \frac{MR^2}{2} \end{bmatrix}}$$

Special cases: centroidal moments of inertia in 2D (with respect to the cm, rotation about $\boldsymbol{e}_3$):

- **Disk / cylinder:** $M_{\mathrm{cm}} = I_{\mathrm{cm}}\dot{\omega}$ with $\boxed{I_{\mathrm{cm}} = \frac{MR^2}{2}}$.
- **Slender rod/bar** (the cylinder with $R \to 0$), rotating about an axis through the cm perpendicular to it: $\boxed{I_{\mathrm{cm}} = \frac{ML^2}{12}}$.

Remark

In these 2D examples we assumed $\dot{I}_{\mathrm{cm}} = 0$. Justification: for rotation about x_3, $M_{\mathrm{cm},3} = \frac{\mathrm{d}}{\mathrm{d}t}(I_{\mathrm{cm},33}\omega_3)$ with $I_{\mathrm{cm},33} = \int_{\mathcal{B}}(\bar{x}_1^2 + \bar{x}_2^2)\rho\,\mathrm{d}V = \int \bar{r}^2\rho\,\mathrm{d}V$, which is constant because in a 2D rotation mass and shape (and the distances $\bar{r}$ to the axis) stay the same. Hence $\boxed{M_{\mathrm{cm}} = I_{\mathrm{cm}}\dot{\omega}}$.

Example: Pulling up a weight

A pulley (mass M, radius R, fixed axle, $I_{\mathrm{cm}} = MR^2/2$) is driven by a torque T; a rope wound on it lifts a weight m. Let S be the rope force, x the height of the weight.

$$\begin{aligned} \text{LMB (weight):} \quad & m\ddot{x} = S - mg, \\ \text{AMB (pulley, about its fixed centre):} \quad & I_{\mathrm{cm}}\ddot{\varphi} = T - SR, \\ \text{kinematics:} \quad & x = R\varphi. \end{aligned}$$

Eliminate $S = \frac{T}{R} - \frac{I_{\mathrm{cm}}}{R}\ddot{\varphi} = \frac{T}{R} - \frac{MR}{2}\frac{\ddot{x}}{R}$:

$$m\ddot{x} = \frac{T}{R} - \frac{M}{2}\ddot{x} - mg \implies \boxed{\ddot{x} = \frac{T/R - mg}{m + M/2}}.$$

The pulley acts like an additional mass $M/2$ that must be accelerated.

4.3 The (mass) moment of inertia tensor

Lecture 11 (21.10.2026)

Recall $\underline{\underline{I}}_B = \int_{\mathcal{B}} [(\boldsymbol{r}\cdot\boldsymbol{r})\underline{\underline{\mathbb{1}}} - \boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}]\rho\,\mathrm{d}V$, with $\boldsymbol{r}$ measured from B.

Additivity. Integrals over a union are sums of integrals, so if $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ (both parts using the *same* reference point B):

$$\underline{\underline{I}}_B = \int_{\mathcal{B}_1} \ldots\,\mathrm{d}V + \int_{\mathcal{B}_2} \ldots\,\mathrm{d}V = \underline{\underline{I}}_{B,1} + \underline{\underline{I}}_{B,2}. \tag{4.13}$$

Holes can be handled by *subtraction*: hollow cylinder = full cylinder (R_o) − inner cylinder (R_i):

$$\underline{\underline{I}}_{\mathrm{cm}} = \underline{\underline{I}}^{(o)}_{\mathrm{cm}} - \underline{\underline{I}}^{(i)}_{\mathrm{cm}}.$$

From Example 3.7:

$$I_{11} = I_{22} = \frac{M_o}{12}(L^2 + 3R_o^2) - \frac{M_i}{12}(L^2 + 3R_i^2), \qquad I_{33} = \frac{M_o R_o^2}{2} - \frac{M_i R_i^2}{2}, \qquad M_{o/i} = \rho\pi R_{o/i}^2 L.$$

Thin ring limit: $R_i = R - t/2$, $R_o = R + t/2$, $t \to 0$: $I_{33} = \frac{\rho\pi L}{2}(R_o^4 - R_i^4) = \frac{\rho\pi L}{2}(R_o^2 + R_i^2)(R_o^2 - R_i^2) \to \underbrace{\rho\, 2\pi R t L}_{M} R^2 = MR^2$ (cf. Example 2.6).

Symmetry axes ($\rho = \mathrm{const}$).

Key result

If the x_i-axis is an axis of rotational symmetry, *or* if the x_i-axis stands normal on a plane of symmetry, then all off-diagonal components with index i vanish. (The reference point must lie on that axis / plane.)

Example: cylinder/cone with apex B, x_1 along the symmetry axis.

- x_1: rotational symmetry $\Rightarrow I_{12} = I_{21} = I_{13} = I_{31} = 0$;
- x_2, x_3: normal to symmetry planes $\Rightarrow I_{23} = I_{32} = 0$.

Why? $I_{12} = -\int_{\mathcal{B}} x_1 x_2 \rho \, \mathrm{d}V = 0$: in any slice $x_1 = \text{const}$ (a circle centred on the axis), for every point with $+x_2$ there is one with $-x_2$, so the contributions cancel. Same argument for $I_{23} = -\int x_2 x_3 \rho \, \mathrm{d}V = 0$.

Positive definiteness and principal axes. For an arbitrary unit vector $\boldsymbol{n}$:

$$\boldsymbol{n} \cdot \underline{\underline{I}}_B \boldsymbol{n} = \int_{\mathcal{B}} [(\boldsymbol{r}\cdot\boldsymbol{r})(\boldsymbol{n}\cdot\boldsymbol{n}) - (\boldsymbol{n}\cdot\boldsymbol{r})(\boldsymbol{r}\cdot\boldsymbol{n})]\rho \, \mathrm{d}V = \int_{\mathcal{B}} [|\boldsymbol{r}|^2 - (\boldsymbol{r}\cdot\boldsymbol{n})^2]\rho \, \mathrm{d}V > 0 \quad \forall \boldsymbol{n},$$

since the projection of $\boldsymbol{r}$ onto $\boldsymbol{n}$ is never longer than $\boldsymbol{r}$ itself ($|\boldsymbol{r}|^2 - (\boldsymbol{r}\cdot\boldsymbol{n})^2$ is the squared distance of the point from the axis $\boldsymbol{n}$).

Key result: Principal axes

$\underline{\underline{I}}_B$ is symmetric and positive definite $\Rightarrow$ it is diagonalisable with real, positive eigenvalues. Hence for every B there exists a frame C such that

$$[\underline{\underline{I}}_B]_C = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix}, \qquad I_1, I_2, I_3 > 0, \tag{4.14}$$

whose axes are the *principal axes* of $\mathcal{B}$ (the eigenvectors). Axes of rotational symmetry and axes normal to symmetry planes are always principal axes.

In practice always aim for a diagonal tensor: then $[\boldsymbol{M}_{\mathrm{cm}}] = [\underline{\underline{I}}_{\mathrm{cm}}][\dot{\boldsymbol{\omega}}] = (I_1\dot{\omega}_1,\, I_2\dot{\omega}_2,\, I_3\dot{\omega}_3)^{\mathsf{T}}$.

Special case 2D. Rotation about $\boldsymbol{e}_3$ through B: AMB $M_B = I_{B,33}\dot{\omega}$ with $I_{B,33} = \int_{\mathcal{B}}(x_1^2 + x_2^2)\rho \, \mathrm{d}V = \int_{\mathcal{B}} r^2 \rho \, \mathrm{d}V$.

Example: Long slender rod about its end

Rod of length L, cross-section A, pivot B at one end, x_1 along the rod. For a slender rod $x_2 \approx 0$ across the cross-section, so

$$I_B = \int_0^L x_1^2 \rho A \, \mathrm{d}x_1 = \frac{1}{3} L^3 \rho A = \boxed{\frac{ML^2}{3}}.$$

Parallel axis theorem (Steiner's theorem).

Key result: Steiner

Let $\Delta\boldsymbol{x} = \boldsymbol{r}_B - \boldsymbol{r}_{\mathrm{cm}} = (\Delta x_1, \Delta x_2, \Delta x_3)$ for a point B. Then

$$[\underline{\underline{I}}_B] = [\underline{\underline{I}}_{\mathrm{cm}}] + M \begin{bmatrix} \Delta x_2^2 + \Delta x_3^2 & -\Delta x_1 \Delta x_2 & -\Delta x_1 \Delta x_3 \\ -\Delta x_1 \Delta x_2 & \Delta x_1^2 + \Delta x_3^2 & -\Delta x_2 \Delta x_3 \\ -\Delta x_1 \Delta x_3 & -\Delta x_2 \Delta x_3 & \Delta x_1^2 + \Delta x_2^2 \end{bmatrix}. \tag{4.15}$$

In 2D: $\boxed{I_B = I_{\mathrm{cm}} + M|\Delta\boldsymbol{x}|^2}$.

Derivation. Write $\boldsymbol{r} - \boldsymbol{r}_B = \boldsymbol{s} - \Delta\boldsymbol{x}$ with $\boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}_{\mathrm{cm}}$. Expanding $(\boldsymbol{s} - \Delta\boldsymbol{x})\cdot(\boldsymbol{s} - \Delta\boldsymbol{x})\underline{\underline{\mathbb{1}}} - (\boldsymbol{s} - \Delta\boldsymbol{x})(\boldsymbol{s} - \Delta\boldsymbol{x})^{\mathsf{T}}$, all terms linear in $\boldsymbol{s}$ integrate to zero because $\int \boldsymbol{s}\,\mathrm{d}m = \boldsymbol{0}$ (definition of the cm). What remains is $\underline{\underline{I}}_{\mathrm{cm}} + M[(\Delta\boldsymbol{x}\cdot\Delta\boldsymbol{x})\underline{\underline{\mathbb{1}}} - \Delta\boldsymbol{x}\Delta\boldsymbol{x}^{\mathsf{T}}]$.

Common pitfall

Steiner always goes *from the centre of mass*: $I_B = I_{\text{cm}} + Md^2$. Shifting between two arbitrary points $A \to B$ requires going through the cm. Consequently I_{cm} is the smallest moment of inertia among all parallel axes.

Steiner examples

1. **Slender rod:** $I_{\text{cm}} = \frac{ML^2}{12}$, $|\Delta x| = \frac{L}{2} \Rightarrow I_B = \frac{ML^2}{12} + M\frac{L^2}{4} = \frac{ML^2}{3}$ ✓ (matches the direct integration).
2. **Rolling wheel:** $I_{\text{cm}} = \frac{MR^2}{2}$; about the contact point A: $I_A = I_{\text{cm}} + MR^2 = \frac{3}{2}MR^2$. The AMB can be used either as $M_{\text{cm}} = I_{\text{cm}}\dot{\omega}$ or $M_A = I_A\dot{\omega}$ (the latter valid for rolling disks, see below).

Relation to area moments of inertia (thin plate, thickness t, x_3 normal to the plate). Recall from Mechanics II (beams):

- polar moment: $I_p = \int_{\mathcal{B}} r^2 \,\mathrm{d}A = \dfrac{I_{33}}{\rho t}$,
- area moments: $I_{11} = \int_{\mathcal{B}}(x_2^2 + x_3^2)\rho \,\mathrm{d}V \approx \int x_2^2 \,\rho t \,\mathrm{d}A = \rho t\, I_x$, $\quad I_{22} = \int(x_1^2 + x_3^2)\rho \,\mathrm{d}V \approx \int x_1^2 \rho t \,\mathrm{d}A = \rho t\, I_y$,
- $I_{12} = -\rho t\, I_{xy}$ (product of area / deviation moment).

So mass moments $= \rho t \times$ area moments for thin plates.

Worked example with several bodies

Lecture 12 (26.10.2026)

Example 3.10: Pulley system

Pulley ① (mass m, radius r, $I_{\text{cm}} = \frac{mr^2}{2}$) is mounted on a fixed axle at the ceiling and driven by a force F acting tangentially on its rim. A rope runs from pulley ① down around the free (hanging) pulley ② (mass M, radius R) and up to a fixed point on the ceiling. A weight ③ of mass m_w hangs from the centre of pulley ②. Rope forces: S_1 (segment between ① and ②), S_2 (segment to the ceiling), S_3 (between ② and ③).

Kinematics. Let x be the upward displacement of the centre of pulley ② (and of the weight). The rope segment going to the ceiling does not move, so the point where it leaves pulley ② is its ICR. Hence $x = R\theta$ and the opposite side of pulley ② moves with $2\dot{x}$ – so the rope (and the rim of pulley ①) moves $2x$:

$$x = R\theta, \qquad r\varphi = 2R\theta = 2x \qquad \Longrightarrow \quad 1 \text{ DOF}.$$

Kinetics.

$$\text{③ LMB:} \quad m_w\ddot{x} = S_3 - m_w g \tag{I}$$

$$\text{① AMB (fixed centre):} \quad I_{\text{cm}}\ddot{\varphi} = (F - S_1)r$$

$$\Rightarrow \frac{mr^2}{2}\cdot\frac{2\ddot{x}}{r} = Fr - S_1 r \;\Rightarrow\; m\ddot{x} = F - S_1 \tag{II}$$

$$\text{② AMB about its ICR:} \quad I_{\text{ICR}}\ddot{\theta} = S_1\, 2R - (S_3 + Mg)R, \quad I_{\text{ICR}} = \tfrac{3}{2}MR^2$$

$$\Rightarrow \tfrac{3}{2}M\ddot{x} = 2S_1 - Mg - S_3 \tag{III}$$

(Taking moments about the ICR eliminates the unknown S_2.) Now 2·(II)+(III)+(I) eliminates S_1 and S_3:

$$2m\ddot{x} + \tfrac{3}{2}M\ddot{x} + m_w\ddot{x} = 2F - Mg - m_w g \implies \boxed{\ddot{x} = \frac{2F - (M+m_w)g}{2m + \frac{3}{2}M + m_w}}.$$

Remark: AMB about the ICR

In general the AMB about a moving point B contains extra terms. For a rolling disk the contact point A has $\boldsymbol{v}_A = \boldsymbol{0}$ instantaneously and its acceleration points towards the cm, which makes $\boldsymbol{M}_A = I_A\dot{\boldsymbol{\omega}}$ valid. Use it for rolling wheels – but do not apply it blindly to arbitrary moving points.

4.3.1 Angular momentum transfer formula

The AMB $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$ is valid for any point B (not necessarily on the body). With $\boldsymbol{H}_B = \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B) \times \boldsymbol{v}\,\mathrm{d}m$ and $\boldsymbol{r}_B = \boldsymbol{r}_{\mathrm{cm}} + \boldsymbol{r}_{\mathrm{cm},B}$:

$$\boldsymbol{H}_B = \underbrace{\int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_{\mathrm{cm}}) \times \boldsymbol{v}\,\mathrm{d}m}_{\boldsymbol{H}_{\mathrm{cm}}} - \boldsymbol{r}_{\mathrm{cm},B} \times \underbrace{\int_{\mathcal{B}} \boldsymbol{v}\,\mathrm{d}m}_{=\boldsymbol{P}} = \boldsymbol{H}_{\mathrm{cm}} + \boldsymbol{P} \times \boldsymbol{r}_{\mathrm{cm},B}.$$

Writing the same for a point A and subtracting ($\boldsymbol{r}_{\mathrm{cm},B} - \boldsymbol{r}_{\mathrm{cm},A} = \boldsymbol{r}_{AB}$):

Key result: Angular momentum transfer formula

$$\boldsymbol{H}_B = \boldsymbol{H}_A + \boldsymbol{P} \times \boldsymbol{r}_{AB}. \tag{4.16}$$

(Analogous to the velocity transfer formula and the moment transfer $\boldsymbol{M}_B = \boldsymbol{M}_A + \boldsymbol{F} \times \boldsymbol{r}_{AB}$ from statics.)

Example I: Disk rolling without slipping, pushed by a force

A disk (mass M, radius R) on rough ground (μ_0) is pulled by a horizontal force F at its centre. Friction F_R acts at the contact point A (backwards).

$$\text{LMB:} \quad Ma_{\mathrm{cm}} = F - F_R,$$

$$\text{AMB about } A: \quad I_A\dot{\omega} = FR, \quad I_A = \tfrac{3}{2}MR^2 \;\Rightarrow\; \dot{\omega} = \frac{2}{3}\frac{F}{MR} \;\Rightarrow\; a_{\mathrm{cm}} = R\dot{\omega} = \frac{2}{3}\frac{F}{M}.$$

Then

$$F_R = F - Ma_{\mathrm{cm}} = \boxed{\tfrac{1}{3}F}.$$

Without friction the disk would just slide (no torque about the cm); the friction force creates the rotation. This is *static* friction (the contact point A is the ICR, $\boldsymbol{v}_A = \boldsymbol{0}$), so it requires $F_R \le \mu_0 N = \mu_0 Mg$, i.e. rolling without slipping is possible for $F \le 3\mu_0 Mg$.

Example II: Spinning disk placed on the ground

A disk spinning with $\omega(0) = \omega_0 > 0$ and $v_{\mathrm{cm}}(0) = 0$ is placed on rough ground ($\mu > 0$). Initially it slips, so *kinetic* friction $F_R = \mu Mg$ acts, in the direction that accelerates the cm.

$$\text{LMB:} \quad Ma_{\mathrm{cm}} = F_R = \mu Mg \;\Rightarrow\; a_{\mathrm{cm}} = \mu g \;\Rightarrow\; v_{\mathrm{cm}}(t) = \mu g t \quad (v_{\mathrm{cm}}(0) = 0),$$

$$\text{AMB about cm:} \quad \frac{MR^2}{2}\dot{\omega} = -F_R R \;\Rightarrow\; \dot{\omega} = -\frac{2\mu g}{R} \;\Rightarrow\; \omega(t) = \omega_0 - \frac{2\mu g}{R}t.$$

Rolling begins when $v_{\mathrm{cm}}(t_f) = R\omega(t_f)$ (then static friction takes over – rolling without slipping, and since no horizontal force is needed to keep rolling at constant speed, $F_R = 0$ afterwards):

$$\mu g t_f = R\Big(\omega_0 - \frac{2\mu g}{R} t_f\Big) \implies \boxed{t_f = \frac{R\omega_0}{3\mu g}}, \qquad v_{\mathrm{cm}}(t_f) = \frac{1}{3} R\omega_0, \qquad \omega(t_f) = \frac{1}{3}\omega_0.$$

Shortcut via angular momentum conservation about the contact point. Let A be the (moving) contact point. The AMB reads $\boldsymbol{M}_A = \dot{\boldsymbol{H}}_A + \boldsymbol{v}_A \times \boldsymbol{P}$; here $\boldsymbol{v}_A \parallel \boldsymbol{P}$ so the last term vanishes, and $\boldsymbol{M}_A = \boldsymbol{0}$ (friction acts at A, gravity and N cancel). Hence $\boldsymbol{H}_A = \text{const}$. Note $\boldsymbol{M}_A = \boldsymbol{0} \neq I_A \dot{\boldsymbol{\omega}}$ here, because $\boldsymbol{v}_A \neq \boldsymbol{0}$ while slipping – use the transfer formula instead:

$$\boldsymbol{H}_A = \boldsymbol{H}_{\mathrm{cm}} + \boldsymbol{P} \times \boldsymbol{r}_{\mathrm{cm},A} = -\frac{MR^2}{2}\omega\, \boldsymbol{e}_3 + (M v_{\mathrm{cm}} \boldsymbol{e}_1) \times (-R\boldsymbol{e}_2) = -\Big(\frac{MR^2}{2}\omega + MR v_{\mathrm{cm}}\Big)\boldsymbol{e}_3 = \text{const}$$

$$\implies \frac{R}{2}\omega + v_{\mathrm{cm}} = \text{const} \implies \frac{R}{2}\omega_0 = \frac{R}{2}\omega(t_f) + v_{\mathrm{cm}}(t_f).$$

At t_f the disk rolls, $R\omega(t_f) = v_{\mathrm{cm}}(t_f)$, so $\frac{R}{2}\omega_0 = \frac{3}{2} v_{\mathrm{cm}}$ and $v_{\mathrm{cm}}(t_f) = \frac{1}{3} R\omega_0$ ✓ – without ever knowing the friction force.

4.3.2 Work–energy balance for rigid bodies

Lecture 13 (28.10.2026)

Kinetic energy of a rigid body. With a reference point C of the body and $\boldsymbol{v} = \boldsymbol{v}_C + \boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C)$:

$$\begin{aligned} T &= \int_{\mathcal{B}} \frac{1}{2} |\boldsymbol{v}|^2 \,\mathrm{d}m = \frac{1}{2} \int_{\mathcal{B}} \left|\boldsymbol{v}_C + \boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C)\right|^2 \mathrm{d}m \\ &= \frac{1}{2} |\boldsymbol{v}_C|^2 \underbrace{\int_{\mathcal{B}} \mathrm{d}m}_{=M} + \boldsymbol{v}_C \cdot \Big(\boldsymbol{\omega} \times \underbrace{\int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_C)\,\mathrm{d}m}_{=M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_C)}\Big) + \frac{1}{2} \int_{\mathcal{B}} \left|\boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C)\right|^2 \mathrm{d}m. \end{aligned}$$

The middle term vanishes if $C = \mathrm{cm}$ (then $\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_C = \boldsymbol{0}$), or if $\boldsymbol{v}_C = \boldsymbol{0}$ (C on the IAR in 3D / the ICR in 2D). For the last term use $(\boldsymbol{\omega} \times \boldsymbol{a}) \cdot (\boldsymbol{\omega} \times \boldsymbol{a}) = [\boldsymbol{a} \times (\boldsymbol{\omega} \times \boldsymbol{a})] \cdot \boldsymbol{\omega}$:

$$\frac{1}{2} \int_{\mathcal{B}} \left[(\boldsymbol{r} - \boldsymbol{r}_C) \times (\boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C))\right] \mathrm{d}m \cdot \boldsymbol{\omega} = \frac{1}{2} (\underline{\underline{I}}_C \boldsymbol{\omega}) \cdot \boldsymbol{\omega}.$$

Key result: Kinetic energy of a rigid body

For $C = \mathrm{cm}$ or $C = \mathrm{ICR/IAR}$ ($\boldsymbol{v}_C = \boldsymbol{0}$):

$$T = \frac{1}{2}\Big[\underbrace{M|\boldsymbol{v}_C|^2}_{\text{translation}} + \underbrace{\boldsymbol{\omega} \cdot \underline{\underline{I}}_C \boldsymbol{\omega}}_{\text{rotation}}\Big], \qquad \text{in 2D: } T = \frac{M}{2} v_C^2 + \frac{1}{2} I_C \omega^2. \tag{4.17}$$

Key result: WEB for rigid bodies

$$T(t_2) - T(t_1) = W_{12} = \sum_i \int_{\boldsymbol{r}_i(t_1)}^{\boldsymbol{r}_i(t_2)} \boldsymbol{F}_i^{\mathrm{ext}} \cdot \mathrm{d}\boldsymbol{r}_i + \int_{t_1}^{t_2} \boldsymbol{M}_T \cdot \boldsymbol{\omega}\,\mathrm{d}t, \tag{4.18}$$

where $\boldsymbol{r}_i$ is the point of attack of the force $\boldsymbol{F}_i$, and external torques $\boldsymbol{M}_T$ also do work. As before, for a conservative system $T + V = \text{const}$.

Example 3.11: Rolling disk rolling up a hill

A disk (mass M, radius R) rolls without slipping with $v_0 = R\omega_0$ onto a slope. Find the maximum height $h_{\max}$.

Conservative: N and the friction force F_R do no work – static friction acts on a stationary point (the ICR) ⇒ no work. So energy conservation applies.

Kinetic energy – two equivalent ways (with $\omega = v_{\mathrm{cm}}/R$):

$$C = \mathrm{cm}: \quad T = \frac{M}{2}v_{\mathrm{cm}}^2 + \frac{1}{2}\underbrace{\frac{MR^2}{2}}_{I_{\mathrm{cm}}}\left(\frac{v_{\mathrm{cm}}}{R}\right)^2 = \frac{M}{2}v_{\mathrm{cm}}^2 + \frac{M}{4}v_{\mathrm{cm}}^2 = \frac{3}{4}Mv_{\mathrm{cm}}^2,$$

$$C = \mathrm{ICR}: \quad T = \frac{M}{2}\underbrace{v_{\mathrm{ICR}}^2}_{=0} + \frac{1}{2}\underbrace{\left(\frac{MR^2}{2} + MR^2\right)}_{I_{\mathrm{ICR}}\ (\mathrm{Steiner})}\omega^2 = \frac{3}{4}Mv_{\mathrm{cm}}^2.$$

WEB:

$$\frac{3}{4}Mv_0^2 = Mgh_{\max} \implies \boxed{h_{\max} = \frac{3}{4}\frac{v_0^2}{g}}.$$

Compare: a particle thrown up with v_0 reaches only $h_{\max} = \frac{1}{2}\frac{v_0^2}{g}$ – a lot of kinetic energy is stored in the rotation.

Example 3.12: Planetary gear

A sun gear (radius r_s, $I_{\mathrm{cm},s}$) driven by a torque M_T meshes with three planet gears (radius r_p), which also mesh with a fixed outer ring. Given M_T, find $\dot{\omega}_s$. (Gravity is neglected because the arrangement is symmetric.)

Forces at the contacts: the internal contact forces between two gears act on both with the same magnitude and opposite direction; since the contact points move together, they do no net work. The ring is fixed and each planet rolls on it, so its contact point with the ring is the ICR of the planet (⇒ no work).

WEB.

$$T(t) = \frac{1}{2}I_{\mathrm{cm},s}\,\omega_s(t)^2 + 3\cdot\frac{1}{2}I_{\mathrm{ICR},p}\,\omega_p(t)^2,$$

($I_{\mathrm{ICR},p} = I_{\mathrm{cm},p} + m_p r_p^2$, assumed constant), and

$$T(t) - T(t_0) = W_{12} = \int_{\varphi(t_0)}^{\varphi(t)} M_T\,\mathrm{d}\varphi_s = M_T\left[\varphi(t) - \varphi(t_0)\right].$$

Differentiate in time:

$$I_{\mathrm{cm},s}\,\omega_s\dot{\omega}_s + 3I_{\mathrm{ICR},p}\,\omega_p\dot{\omega}_p = M_T\,\omega_s.$$

Kinematics (Mechanics I): the sun–planet contact point has speed $\omega_s r_s$; for the planet this point is at distance $2r_p$ from its ICR, so $\omega_s r_s = \omega_p\,2r_p \Rightarrow \omega_p = \dfrac{\omega_s r_s}{2r_p}$. Insert:

$$I_{\mathrm{cm},s}\,\omega_s\dot{\omega}_s + 3I_{\mathrm{ICR},p}\frac{\omega_s r_s}{2r_p}\cdot\frac{\dot{\omega}_s r_s}{2r_p} = M_T\omega_s \implies \boxed{\dot{\omega}_s = \frac{M_T}{I_{\mathrm{cm},s} + 3I_{\mathrm{ICR},p}\left(\dfrac{r_s}{2r_p}\right)^2}}.$$

The planets act like an extra ("reduced") moment of inertia on the sun shaft.

Example 3.13: A note on gears

Gear 1 (radius r_1, I_1, driving torque M_1) drives gear 2 (radius r_2, I_2, load torque M_2); the tooth force is F_r.
Kinematics (rolling without slipping at the contact): $r_1\omega_1 = r_2\omega_2 \Rightarrow \dfrac{\omega_1}{\omega_2} = \dfrac{r_2}{r_1}$.
AMB about the fixed centres:

$$I_1\dot{\omega}_1 = M_1 - F_r r_1, \qquad I_2\dot{\omega}_2 = F_r r_2 - M_2.$$

Eliminate $F_r = (M_1 - I_1\dot{\omega}_1)/r_1$:

$$M_2 = M_1\frac{r_2}{r_1} - \underbrace{\left(I_1\dot{\omega}_1\frac{r_2}{r_1} + I_2\dot{\omega}_2\right)}_{\text{losses because of inertia}}.$$

If the gears are massless ($I_1 = I_2 = 0$):

$$\boxed{\frac{M_1}{M_2} = \frac{r_1}{r_2}} \quad\Longrightarrow\quad \boxed{M_1\omega_1 = M_2\omega_2}$$

– the transmitted power is conserved: a gear trades torque for angular velocity.

Formula Summary

Topic	Formula
Polar kinematics	$\boldsymbol{v} = \dot{r}\boldsymbol{e}_r + r\dot{\varphi}\boldsymbol{e}_\varphi$, $\boldsymbol{a} = (\ddot{r} - r\dot{\varphi}^2)\boldsymbol{e}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi})\boldsymbol{e}_\varphi$
Path coordinates	$\boldsymbol{v} = \dot{s}\boldsymbol{e}_t$, $\boldsymbol{a} = \ddot{s}\boldsymbol{e}_t + \frac{v^2}{\rho}\boldsymbol{e}_n$
LMB (particle/system/body)	$\sum \boldsymbol{F}^{\text{ext}} = \dot{\boldsymbol{P}}$, $\boldsymbol{P} = M\boldsymbol{v}_{\text{cm}}$; const. mass: $\sum \boldsymbol{F}^{\text{ext}} = M\boldsymbol{a}_{\text{cm}}$
Variable mass	$m\boldsymbol{a} = \sum \boldsymbol{F} + \dot{m}\boldsymbol{v}_{\text{rel}}$, $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v}_m - \boldsymbol{v}$
AMB	$\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$; B = cm or fixed: $\boldsymbol{M}_B = \frac{\mathrm{d}}{\mathrm{d}t}(\underline{\underline{I}}_B\boldsymbol{\omega})$; 2D: $M_B = I_B\dot{\omega}$
WEB	$T_2 - T_1 = W_{12}$; $T_2 + V_2 - (T_1 + V_1) = W_{12}^{\text{non-cons.}}$; $\boldsymbol{F} = -\nabla V$
Rigid body KE	$T = \frac{1}{2}Mv_C^2 + \frac{1}{2}\boldsymbol{\omega}\cdot\underline{\underline{I}}_C\boldsymbol{\omega}$ (C = cm or $\boldsymbol{v}_C = \boldsymbol{0}$)
Impact	v_t unchanged; $e = \hat{F}_{\text{rest}}/\hat{F}_{\text{comp}}$; wall: $v_n^+ = -ev_n^-$
Collision	$e = -\dfrac{v_2^{n+} - v_1^{n+}}{v_2^{n-} - v_1^{n-}}$, momentum conserved
Velocity transfer	$\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega}\times\boldsymbol{r}_{AB}$
Acceleration transfer	$\boldsymbol{a}_B = \boldsymbol{a}_A + \dot{\boldsymbol{\omega}}\times\boldsymbol{r}_{AB} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}_{AB})$
Projected velocities	$\boldsymbol{v}_A\cdot\boldsymbol{r}_{AB} = \boldsymbol{v}_B\cdot\boldsymbol{r}_{AB}$
Angular momentum	$\boldsymbol{H}_B = M(\boldsymbol{r}_{\text{cm}} - \boldsymbol{r}_B)\times\boldsymbol{v}_B + \underline{\underline{I}}_B\boldsymbol{\omega}$; $\boldsymbol{H}_B = \boldsymbol{H}_A + \boldsymbol{P}\times\boldsymbol{r}_{AB}$
Inertia tensor	$\underline{\underline{I}}_B = \int[(\bar{\boldsymbol{r}}\cdot\bar{\boldsymbol{r}})\underline{\underline{\mathbb{1}}} - \bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}}]\,\mathrm{d}m$
Steiner	$I_B = I_{\text{cm}} + Md^2$
Standard values	disk $\frac{MR^2}{2}$, ring MR^2, rod (cm) $\frac{ML^2}{12}$, rod (end) $\frac{ML^2}{3}$, cylinder $I_{11} = \frac{MR^2}{4} + \frac{ML^2}{12}$
DOFs	particle: $d - k$; n particles: $nd - r$; n rigid bodies: $n\frac{d(d+1)}{2} - k$